MILITARY SPECIFICATION

SHELTER, ELECTRICAL EQUIPMENT
S-538( )/TSQ-84

This specification is approved for use by all Departments and Agencies of the DOD.

1. SCOPE

1.1 Scope.- This specification covers the air or vehicular transportable shelter facility part of Communications Technical Control Center AN/TSQ-84( ) which has provisions for 432 four wire circuits in and 432 out, hard wired normal through; and 24 four wire circuits capable of switching from binding post to 26 pair cable. An additional six signal connectors are provided in the rear signal entrance box for alarm orderwires and control system. Forty-eight 2-wire attenuators, eighteen 2-wire amplifiers and nine 2-wire amplitude equalizers located on the MDF patch panels provide facilities for conditioning of analog circuits. (See 6.1 and 6.5)

1.2 Description.- Shelter Electrical Equipment S-538( )/TSQ-84 hereinafter called the shelter facility, includes a modified S-280( )/G shelter and all patch panels, racks, brackets, shelves, tie downs, electrical components and wiring to enable installation and operation of assemblage components of the AN/TSQ-84( ). (See 6.1 and 6.2)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment)

FSC 5410

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MIL-W-6858	Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys & Titanium, Spot & Seam
MIL-T-7928	Terminals, Lug & Splice, Crimp Style Copper
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-W-45205	Welding, Gas, Metal-Arc & Gas, Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-S-55505	Shelter, Electrical Equipment (Containing Equipment) Packaging & Packing of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual & Mechanical Defects
MIL-STD-781	Reliability Tests: Expotential Distribution
MIL-STD-785	Reliability Program for Systems and Equipments: Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-1235	Single and Multi-level Continuous Sampling Procedures & Tables for Inspection by Attributes
MIL-STD-1474	Noise Limits for Army Material

DRAWINGS

ELECTRONICS COMMAND

DL-SC-B-608368	Shelter, Electrical Equipment S-538( )/TSQ-84

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SC-F-608401 Power Schematic Diagram S-538( )/TSQ-84

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 First Article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.- The shelter facility shall be constructed in accordance with drawings listed on DL-SC-B-608368. (See 4.6)

3.3 Electrical Installation.- (See 3.22)

3.3.1 Power system wiring and components.- (See 4.7.1) Installation and wiring of power wires and components (including cable assemblies, receptacles, switches, meters, circuit breakers, neon lamps, fluorescent lamps, incandescent lamps, lighting fixtures, distribution assembly system and wiring components) constituting a four-wire (115/230 VAC plus ground single phase, 60 hz, electrical power distribution system) shall be in accordance with the drawings. The completed in-place power distribution system shall meet the following requirements:

3.3.1.1 Continuity of conductors.- Electrical continuity shall exist between each pin of the power entrance connectors and respective circuit termination thereof when tested in accordance with 4.7.1 and 4.7.1.1. The continuity resistance shall be less than 1 ohm.

3.3.1.2 Dielectric breakdown voltage.- The insulation of all power wires, cables, and components shall withstand a potential of 500 volts DC applied continuously for not less than 30 seconds when tested in accordance with 4.7.1 and 4.7.1.2.

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3.3.1.3 Insulation resistance.- The resistance of insulation between any two circuit conductors and also between each of the three ungrounded conductors and shelter facility ground terminal shall not be less than 100 megohms when tested in accordance with 4.7.1 and 4.7.1.3.

3.3.2 Power distributional and operational functions tests.- Main power branch distribution and circuit switching functions thereof shall meet the operational requirements as denoted by the Power Schematic Drawing SC-F-608401 when tested in accordance with 4.7.2.

3.3.3 Signal wiring and components.- (See 4.7.3) Installation and wiring of signal wires, cables and wiring components (including binding posts, connectors, plugs, jacks, switches, terminal strips, terminal blocks, lightning arrestors, and wiring devices) shall be in accordance with the drawings. The completed in-place signal installation system shall meet the following requirements:

3.3.3.1 Continuity resistance of conductors.- Circuit continuity resistance of all signal lines shall meet the following requirements when tested in accordance with 4.7.3 and 4.7.3.1.

a. Less than 5 ohms continuity resistance on normal thru lines.

b. 3 kilo-ohms $\pm$ 10% VF line terminations.

c. 620 ohms $\pm$ 10% idle line resistance on all VF lines.

3.3.3.2 Dielectric strength and insulation resistance tests.- The insulation of all signal wires, cables and components shall withstand a potential of 500 volts DC applied continuously for not less than 5 seconds across each normal thru line and shelter facility ground. The insulation resistance shall not be less than 100 megohms. The test method shall be in accordance with 4.7.3.2.

3.3.4 Engineering transmission tests.- (See 4.7.4)

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3.3.4.1 Insertion loss/frequency response.-

a. The insertion loss at 1 Khz of any VF line shall not be greater than 0.5 db when tested in accordance with 4.7.4.1.

b. The frequency response at the specified frequencies shall be within ± 1 db of the 1 Khz level when tested in accordance with 4.7.4.1.

3.3.4.2 Crosstalk.- Crosstalk couplings between any two circuits terminated on each side in a 600 ohm impedance shall not be less than 60 db at frequencies up to 32.0 Khz when tested in accordance with 4.7.4.2.

3.3.4.3 Longitudinal balance.- The longitudinal balance of any VF line shall not be less than 40 db at the specified frequencies when tested in accordance with 4.7.4.3.

3.3.4.4 Impedance.- The input and output impedance of any VF line shall be 604 ohms + 10% at the specified frequencies when tested in accordance with 4.7.4.4.

3.3.5 Test of line conditioning components.-

3.3.5.1 Line amplifiers tests.-

a. Gain and output power - The gain of each amplifier shall be capable of being adjusted from zero db to 28 db minimum (at maximum gain setting) at 1 Khz, with a maximum power output of 9 dbm when tested in accordance with 4.7.5.1a.

b. Distortion - The distortion of each amplifier shall be 3% or less at the specified frequencies when tested in accordance with 4.7.5.1b.

c. Frequency response - The output level shall not change more than + 0.30 or - 0.40 db from the 1 Khz power output level between the frequency range of 300 to 3400 hz when tested in accordance with 4.7.5.1c.

d. Noise - The noise level introduced by each amplifier shall not exceed - 65 dbm when tested in accordance with 4.7.5.1d.

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3.3.5.2 Attenuators tests.- Each attenuator shall be tested in accordance with 4.7.5.2 and meet the following requirements:

a. With the attenuator's switch fully clockwise and an input signal of 0 db the voltmeter reading across each attenuator output jack shall be 0 db ± 0.20 db.

b. With the attenuator's switch rotated counterclockwise, one step at a time, the voltmeter readings shall increase respectively by 1 db increments, up to a maximum of 10 db. Each reading shall be ± 0.20 db of the required level.

3.3.5.3 Line equalizers tests.- Each line equalizer shall be capable of providing an adjustable compensation of 0 to 17 db maximum between 300 hz and 3 Khz when tested in accordance with 4.7.5.3.

3.3.6 Cable assemblies, power and audio frequency.- All power and audio frequency cable and cord assemblies shall be fabricated in accordance with the drawings and meet the continuity, dielectric, insulation resistance, and pull tests requirements of MIL-C-3885. (See 4.4)

3.4 Operational tests.- With all operational equipments of the AN/TSQ-84 assemblage installed, the operational requirements of all power and audio cable and cord assemblies, switches, connectors, and jack assemblies shall be met when tested in accordance with 4.8.

3.5 Vehicular transportation.- The shelter facility shall be capable of being transported over cross-country terrain by military vehicles without sustaining any permanent damage such as buckling, cracking, delamination, or other permanent deformation. (See 4.9)

3.6 Rail transportation.- The shelter facility, without any extra tying, taping, etc., shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail and shall be subjected to withstand the test of paragraph 4.10 without sustaining any buckling, cracking, delamination, or other permanent deformation.

3.7 Drops, flat and rotational.- The shelter facility shall be capable of withstanding drops of 18 inches onto concrete. After all drop tests of 4.11.1 and 4.11.2 there shall be no buckling, cracking, delamination or other permanent deformation.

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3.8 Watertightness.- The shelter facility shall be capable of withstanding the test of 4.12. During and after the test there shall be no leakage into the shelter facility or its walls, floor, roof, or door.

3.9 Fording.- The shelter facility shall be capable of withstanding the tests of 4.13. During and after the test there shall be no leakage into the shelter facility or its walls, floor, or door.

3.10 Environmental, shelter assemblage.- (See 6.8)

3.10.1 Temperature extremes.- The assemblage shall give full operation at all ambient temperatures and extremes of - 25°F to + 125°F. This shall include accessories such as heating and cooling system, electronic equipment temperature control system and all external components which are exposed to the ambient conditions during operation. (See 4.14.1)

3.10.1.1 Temperature affecting electronic equipment mounted in shelter.- Full operation, except for teletypewriters and line amplifiers, shall start after two hours of warmup or cooling by the environmental control equipment. The low temperature operating limit of the teletypewriters and line equalizers is +32°F. (See 4.14.1)

3.10.1.2 Temperature non-operating (storage and transit).- The assemblage shall be capable of withstanding a high storage temperature of + 160°F for a period of 48 hours and a low storage temperature of - 70°F for a period of 24 hours without deterioration or change in tolerance limits which could prevent it from meeting operational, service and maintenance requirements. (See 4.14.1.2 and 4.14.1.3)

3.10.2 Humidity, shelter assemblage.- The assemblage shall be capable of withstanding the humidity tests specified in Method 507, Procedure II, of MIL-STD-810 without corrosion, deterioration, or change in tolerance which could prevent it from meeting operational, service and maintenance requirements. (See 4.14.1.4)

3.11 Humidity, hardware.- Hardware sample shall have no evidence of corrosion, rust or deterioration after testing. (See 4.15)

3.12 Mounting inserts.- All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive, and when installed, each insert shall be capable of withstanding the

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tensile load and the torque load referenced in 4.16. There shall be no damage or permanent distortion to the shelter facility mounting surface or insert head or threads and the insert shall not have rotated from its original position. Insert threads shall be free of paint.

3.13 Acoustical noise.- With the air conditioners operating the acoustical noise level at the operator's positions shall not exceed the NC-60 criteria as specified in MIL-STD-1474(MI), Table 3, Catagory F. (See 4.17)

3.14 Cleaning.- (See 3.24)

3.14.1 Parts.- After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.14.2 Units.- After assembly, the equipment shall be cleaned thoroughly and shall be free from excess solder or flux, loose hardware, and other foreign material.

3.15 Finish.- The shelter facility shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.16 Marking.- (See 3.24 and 4.4)

3.16.1 General.- Marking shall conform to Specification MIL-M-13231 and the drawings.

3.16.2 Nameplates.- Each shelter facility shall be provided with a nameplate in accordance with the drawings. Each nameplate shall be imprinted with a sequential serial number.

3.16.3 Aircraft loading data plate.- The shelter facility shall be provided with data plates in accordance with the drawings, imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage;

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and the bearing pressure on the bottom of the skids. The shelter assemblage is the shelter facility with operational equipments or dummy loads in place. The locations of the centers-of-gravity shall be determined by the contractor using the first article.

3.16.4 Interior and exterior marking.- All interior and exterior marking of the shelter facility shall be as depicted on the drawings.

3.17 Riveting.- (See 3.24) Riveted joints shall be tight, the jointed parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.18 Seal, vapor.- (See 3.24) All joints and raw edges of the framing for the power signal, entrance boxes, and air conditioners shall be sealed to prevent entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners shall be dipped in sealer prior to installation. The sealer shall be one part Silicone Construction Sealant SE-1204 as made by General Electric Company or Silastic 732 as made by Dow Corning, or equal, using the manufacturer recommended primer. Internal sealer shall be Silastic 738 as made by Dow Corning, or equal.

3.19 Securing of parts.- (See 3.24) Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the facility securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified drop and transport tests.

3.19.1 Screws, threaded devices, and related parts.- Screws, threaded devices and related parts shall be as defined by the drawings. Thread forming screws shall not be used.

3.20 Soldering.- (See 3.24)

3.20.1 Solder and flux.- Solder and flux shall be as defined on the drawings.

3.20.2 Acid or acid salts.- No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts

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be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.20.3 Process.- There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, over-heated or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.21 Welding.- Welded joints shall be as defined by the drawings and shall be such that grinding on the finish weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 3.24 and 4.4)

3.22 Wiring and cabling.- (See 3.24) Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

3.22.1 Slack.- Wires and cables shall be as short as practical except that sufficient slack shall be provided:

a. To prevent undue stress on cable forms, wires, and connections.

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b. To facilitate field repair of broken or cut wires.

c. To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.22.2 Protection.- Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.22.3 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation unless specifically indicated by the drawings.

3.22.4 Solderless terminal lug connections.- Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements of MIL-T-7928. (See 4.18)

3.22.5 Identification of wiring.- Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.23 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.19) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portion of the following paragraphs: (See 4.20)

3.2 Construction
3.3 Electrical installation
3.12 Mounting inserts
3.14 Cleaning
3.15 Finish
3.16 Marking
3.17 Riveting
3.18 Seal, vapor
3.19 Securing of parts
3.20 Soldering
3.21 Welding
3.22 Wiring and cabling

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3.25 Reliability.- (See 4.23)

3.25.1 Reliability requirement.- The reliability of the patch panel shall be controlled by use of the best engineering practices and by use of quality components. The failure rate of each segment of wire, including its terminations to connectors and/or jacks, shall be no greater than .02 failures per million hours. The failure rate of each jack shall be no greater than .04 failures per million hours or .0000004 failures per mating.

3.25.2 Reliability program.- The contractor shall conduct a reliability program that shall include, but not limited to, the following reliability tasks as described in MIL-STD-785:

a. Reliability management (5.1)

b. Parts reliability (5.2.3)

c. Failure data (5.4)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform such inspections that are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include preparation for delivery). (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

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(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.21)

4.3 First Article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor. (See 3.1)

4.3.1 First Article units.- The contractor shall furnish one (1) first article unit of the complete S-538( )/TSQ-84.

4.3.2 First Article inspection.- The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.3.3 First Article data.- The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I - First Article Inspection

Inspections		Rqr Para	Test Para
1.	Inspection covered by subsidiary documents	-	4.4
2.	Interchangeability	3.23	4.19
3.	Mounting Inserts	3.12	4.16.1
4.	Solderless-terminal-lug connectors	3.22.4	4.18
5.	Visual and mechanical	3.24	4.20
6.	Power system wiring and components	3.3.1	4.7.1
7.	Power distributional and operational functions tests	3.3.2	4.7.2
8.	Signal wiring and components	3.3.3	4.7.3
9.	Engineering transmission tests	3.3.4	4.7.4
10.	Test of line conditioning components	3.3.5	4.7.5
11.	Operational tests	3.4	4.8

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TABLE I - First Article Inspection (contd)

Inspections	Rqr Para	Test Para
12. Vehicular transportation	3.5	4.9
13. Rail transportation	3.6	4.10
14. Flat drop	3.7	4.11.1
15. Rotational drop	3.7	4.11.2
16. Watertightness	3.8	4.12
17. Fording	3.9	4.13
18. Acoustical noise	3.13	4.17
19. Repeat 6. Power system wiring and components	3.3.1	4.7.1
20. Repeat 7. Power distributional and operational functions tests	3.3.2	4.7.2
21. Repeat 8. Signal wiring and components	3.3.3	4.7.3
22. Repeat 10. Test of line conditioning components	3.3.5	4.7.5
23. Repeat 11. Operational tests	3.4	4.8
24. Environmental, Shelter assemblage	3.10	4.14

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Cable assemblies	3.3.6
Finish	3.15
Marking	3.16
Welding	3.21
Reliability	3.25
Solderless terminal lug connectors	3.22.4

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsiblity for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Lots shall be formed only from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table II.

TABLE II - Group A Inspection

Inspection	Rqr Para	Insp Para	AQL % DEF Major	Minor
Visual and Mechanical	3.24	4.20	1.0	4.0
Electrical installation				**
Power system wiring and components				
Continuity of conductors	3.3.1.1	4.7.1.1	1.0 for	
Dielectric breakdown voltage	3.3.1.2	4.7.1.2	the entire	
Insulation resistance	3.3.1.3	4.7.1.3	electrical	
Power distributional and operational functions tests	3.3.2	4.7.2 thru 4.7.2.2	installation	
Signal wiring and components				
Continuity resistance of conductors	3.3.3.1	4.7.3.1		
Dielectric strength and insulation resistance tests	3.3.3.2	4.7.3.2		
Engineering Transmission test				
Crosstalk	3.3.4.2	4.7.4.2		

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TABLE II - Group A Inspection
(contd)

Inspection	Rqr Para	Insp Para	AQL % DEF Major	Minor
Test of line conditioning components				
Line amplifiers test	3.3.5.1	4.7.5.1		
Attenuators test	3.3.5.2	4.7.5.2		
Line equalizer test	3.3.5.3	4.7.5.3		
Operational check				
Hybrid lines	3.4	4.8.2		
Test of amplifier-headset AM-6789/G	3.4	4.8.4		

** All defects are major

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in the order listed in Table II.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection except for Interchangeability and Mounting Insert Inspections which are performed prior to Group A. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III - Group B Inspection

Inspection	Rqr Para	Insp Para	AQL % DEF (See 4.5.2.1)
Interchangeability	3.23	4.19	4.0
Mounting Insert	3.12	4.16.2	4.0
Watertightness	3.8	4.12	2.5
Fording	3.9	4.13	2.5

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NOTE: The order of inspection of Group B with respect to Groups A and C shall conform to the order specified in 4.3. The order of inspection within Group B shall be as in 4.5.2.2.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table III shall be as follows:

AQL	Inspection level
4.0%	S-4
2.5%	S-5

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed on the same equipment in the order in which they appear.

4.5.3 Group C inspection.- This inspection shall be as listed in Table IV. Samples shall be selected in accordance with 4.5.3.1 thru 4.5.3.4.

TABLE IV - Group C Inspection

Inspection	Rqr Para	Insp Para
Construction verification	3.2	4.6
Solderless-terminal-lug connection	3.22.4	4.18
Mounting inserts	3.12	4.16.1
Acoustical noise	3.13	4.17
Operational tests	3.4	4.8
Vehicular transportation	3.5	4.9
Rail transportation	3.6	4.10
Drops, Flat and Rotational	3.7	4.11

NOTE: The order of inspection of Group C with respect to Groups A and B shall conform to the order specified in 4.3. The order of inspection within Group C for all inspections listed shall be as listed in Table IV. Units subjected to Group C drop tests shall subsequently be subjected to and meet the watertightness and fording tests of 4.12 and 4.13.

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4.5.3.1 Sampling for inspection of construction evaluation.- The Equipment Verification Review, in accordance with paragraph 4.6, shall be performed once on each part and assembly, randomly selected, and on their installation into one of first 10 shelter facilities produced. This inspection shall be performed as the selected unit is being assembled.

4.5.3.2 Sampling for inspection of solderless-terminal-lug-connections.- Five specimens of each combination of wire and terminal produced each week shall be prepared for inspection. A specimen is a one foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

4.5.3.3 Sampling for inspection of drop tests.- One sample, representative of the first 10 units or fraction thereof produced, shall be subjected to the drop tests. This sample shall be the same unit previously subjected to the vehicular and rail transportation tests of 4.5.3.4. Thereafter, one sample from each additional 150 units or fraction thereof produced shall be subjected to the drop tests.

4.5.3.4 Sampling for other inspection.- For inspection tests listed below, one unit shall be selected at random from the first 10 units or fraction thereof which have been produced, unless otherwise specified in the contract. Selection of the unit will be made by the government.

a. Mounting inserts
b. Operational tests
c. Truck transportability
d. Railroad transportability
e. Acoustical noise

4.5.3.5 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified sample units which have been subjected to and passed Group C inspection may be accepted on contract, provided that they are resubjected to and pass Group A inspection after repair of all visible damage. (See 6.4)

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4.5.5 Alternate Group A audit and Group B inspection procedures.- The contractor may elect to use a continuous sampling and inspection plan instead of the lot inspection plans from MIL-STD-105. If this alternate is selected the procedure of implementation must be in accordance with continuous sampling multilevel plans described in MIL-STD-1235. Sampling plans for the AQL's listed in Table II Group A inspection, and Table III Group B inspection of this specification shall be selected from Table VIII, IX and X of MIL-STD-1235 which are for f=1/2. The number of units in the Production Interval shall be the entire production quantity unless otherwise specified in the contract. Before starting production contractor shall notify the contracting officer in writing which method he intends to use. The method may not be changed during production without written approval of the government.

4.6 Construction verification.- The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment order against the drawings (See 3.2) cited in the Technical Data Package (as supplemented by all approved engineering changes). The EVR shall consist of the following:

a. An audit to establish that "as-built" shelter facilities, including individual parts, their use in subassemblies, and assemblies (components), and their final installation into the shelter forming the shelter facility, are in accordance with the end product drawings.

b. An audit to establish that "as-built" shelter facilities, including parts, components and their installation, meet the acceptance requirements specified in the Technical Data Package for each shelter part, components, and shelter facility. The audit shall record all acceptance test methods used together with resulting verification test data.

c. Preparation of EVR report(s) containing the following information:

(1) Identification of parts, components, shelter facilities and details proving compliance with 4.6a and b above.

(2) Discrepancies noted.

(3) Corrective action taken.

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d. Two copies of the EVR report shall be submitted no later than 15 days prior to the submission of the first production unit or lot to the government technical organization cited in the contract. The results of the EVR shall be subject to verification by authorized government personnel at the time the first unit or production lot is offered for acceptance. Government verification will be to the depth and extent necessary to demonstrate that the "as-built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to government personnel during the verification. The contractor shall provide the following:

(1) Segregation of parts, components of shelter facilities, to permit reviewing government personnel access for detailed inspection. The contractor shall also provide personnel to disassemble any parts of the shelter facility necessary to permit government review of the shelter facility, its components and their installation, and to reassemble parts after inspection.

(2) Have as required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

4.7 Electrical installation tests.- The tests of 4.7.1 through 4.7.3.2 shall be performed on each shelter facility. Before applying power to the shelter facility an electrical ground at the power entrance box shall be made. Power to be supplied shall be 115/230 volts AC, 60 hertz, single phase.

4.7.1 Power system wiring and components.- (See 3.3.1) Prior to conducting power wiring and components tests of 4.7.1.1, 4.7.1.2, and 4.7.1.3, the following shall be accomplished. (Refer to power schematic drawing).

a. Remove or disconnect the following electrical loads:

(1) All fluorescent, incandescent, and neon (power indicator and power distribution box) lamps.

(2) All fluorescent lamp starters.

(3) Air conditioners.

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b. Remove power distribution box cover panel and disconnect both conductors from the voltmeter terminals. Position or isolate disconnected leads so as not to short to each other or to shelter facility ground.

4.7.1.1 Continuity of conductors.- Perform the following continuity tests using a calibrated low resistance reading ohmmeter or equivalent.

a. Place the following controls in the ON position:

(1) Fluorescent light switches.

(2) Incandescent cold start switch.

(3) Main circuit breaker.

(4) All branch breakers.

b. Place Blackout/Bypass switch to the Bypass position.

c. Measure continuity resistance as follows. The requirements of 3.3.1.1 shall be met.

	FROM	TO
(1)	Pin 1 of J1 (Power input connector)	Grounding contacts of all power receptacles
(2)	Pin 1 of J1	Main ground terminal
(3)	Pin 1 of J1	Ground terminal in power distribution box
(4)	Pin 2 of J1	Brass contacts of all associated power receptacles as depicted on the drawing.
(5)	Pin 3 of J1	Silver contacts of all power receptacles.
(6)	Pin 3 of J1	White wire terminations at each incandescent and fluorescent lamp.
(7)	Pin 4 of J1	Brass contacts of all associated power receptacles as depicted on the drawing.

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	FROM	TO
(8)	Pin 4 of J1	Black wire terminations at each fluorescent and incandescent lamp.

4.7.1.2 Dielectric breakdown voltage.- Using a non-destructive dielectric tester, apply a test voltage of 500 volts DC for 30 seconds across each conductor under test with all other conductors at ground potential. The requirements of 3.3.1.2 shall be met. The test shall be conducted at the power input connector J1 as follows: (NOTE: To eliminate danger of shock.- Discharge conductors following each test.)

a. Pin 2 with Pins 1, 3 and 4 grounded.

b. Pin 3 with Pins 1, 2 and 4 grounded.

c. Pin 4 with Pins 1, 2 and 3 grounded.

4.7.1.3 Insulation resistance.- Test conditions, preparation, points of measurement and grounding of conductors shall be as given for 4.7.1.2 above. The insulation resistance of all conductors with respect to ground shall meet the requirements of 3.3.1.3. (NOTE: To eliminate danger of shock.- Discharge conductors following each test.)

4.7.2 Power distributional and operational function tests.- Power distributional and operational function tests shall be made subsequent to the tests of 4.7.1 and with all electrical sources and loads connected. All components which are part of and/or associated with the electrical power system shall be in-place and connected. Prior to conducting the tests of 4.7.2.1 and 4.7.2.2, the following shall be accomplished (Refer to power schematic drawing):

a. Retain open shelter door.

b. Place all circuit breakers (including main) to the OFF position.

c. Provide and place suitable test lamps or equivalent test indicators in each power receptacle.

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d. Place all lighting control switches in the ON position.

e. Place the Blackout/Bypass switch in the Bypass position.

4.7.2.1 Test of line voltmeter and indicator lamp.-

a. Apply 115/230 volts AC, single phase, 60 hertz, to shelter facility at power connector J1. Power indicator neon lamps shall light indicating power to the shelter facility.

b. Voltmeter shall indicate nominal 115 volts AC with the line switch in either L1 to N or L2 to N.

c. With the line switch in the L1 to L2 position, the voltmeter shall indicate nominal 230 volts AC.

d. Place main circuit breaker in the ON position.

4.7.2.2 Test of circuit breakers and switches.-

Circuit #1.- (Lights circuit breaker CB1)

a. Place circuit breaker CB1 in the ON position. Neon lamp DS1 shall light.

b. Place LIGHT - 1 switch in the ON position. Half of the roadside and curbside fluorescent lamps shall light.

c. Place LIGHT - 2 switch in the ON position. Remaining half of the roadside and curbside fluorescent lamps shall light.

d. Place console light switch in the ON position. All console incandescent lamps shall light.

e. Place incandescent cold start switch in the ON position. All cold start incandescent lamps shall light.

f. With shelter door open, place blackout/bypass switch in the blackout position. All fluorescent and incandescent lamps shall extinguish.

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g. Close shelter door. All fluorescent and incandescent lamps shall light.

h. Open shelter door and replace blackout/bypass switch to the bypass position to restore lights.

i. Place circuit breaker CB1 in the OFF position. All lights shall extinguish. Restore lights by placing CB1 in the ON position.

Circuit #2.- (Heater - 3 circuit breaker CB2)

a. Place circuit breaker CB2 in the ON position. Neon lamp DS2 and test lamp in heater - 3 receptacle shall light.

b. Place CB2 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #3.- (Heater - 1 circuit breaker CB3)

a. Place circuit breaker CB3 in the ON position. Neon lamp DS3 and test lamp in heater - 1 receptacle shall light.

b. Place CB3 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #4.- (Heater - 2 circuit breaker CB4)

a. Place circuit breaker CB4 in the ON position. Neon lamp DS4 and test lamp in heater - 2 receptacle shall light.

b. Place CB4 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #5.- (DC Power circuit breaker CB5)

a. Place circuit breaker CB5 in the ON position. Neon lamp DS5 shall light.

b. Place 24 volt DC power supply #1 switch in the ON position. Place a DC voltmeter across pins 4 and 5 of each remote/alarm unit mating power connector. The voltmeter shall indicate 24 VDC in each instance.

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c. Place 24 volt DC power supply switch in the OFF position. The voltmeter shall indicate 0 volts for the same meter connections of step b above.

Circuit #6.- (Spare circuit breaker CB6)

a. Place circuit breaker CB6 in the ON position. Neon lamp DS6 shall light.

b. Place circuit breaker CB6 in the OFF position. Neon lamp DS6 shall extinguish.

Circuit #7.- (Spare 220 VAC circuit breaker CB7)

a. Place circuit breaker CB7 in the ON position. Both power indicator neon lamps associated with CB7 shall light.

b. Place an AC voltmeter across contacts X and Z of the spare 220 VAC receptacle. Voltmeter shall indicate approximately 220 VAC.

c. Place voltmeter across contacts X and Y and then across contacts Z and Y of the same receptacle. Voltmeter shall indicate approximately 115 VAC in each instance.

d. Place CB7 in the OFF position. Neon lamps mentioned in step a above shall extinguish. Voltmeter readings of steps b and c above shall indicate 0 volts.

Circuit #8.- (Intercom circuit breaker CB8)

a. Place circuit breaker CB8 in the ON position. Neon lamp DS9 and test lamps in the intercom receptacles shall light.

b. Place circuit breaker CB8 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #9.- (Spare circuit breaker CB9)

a. Place circuit breaker CB9 in the ON position. Neon lamp DS10 shall light.

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b. Place circuit breaker CB9 in the OFF position. Neon lamp DS10 shall extinguish.

Circuit #10.- (TTY/TH-22/TA-248 circuit breaker CB10)

a. Place circuit breaker CB10 in the ON position. Neon lamp DS11 and test lamps in the teletype receptacles, TH-22 receptacle, and telering receptacle shall light.

b. Place circuit breaker CB10 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #11.- (Test equipment circuit breaker CB11)

a. Place circuit breaker CB11 in the ON position. Neon lamp DS12 and test lamps in all equipment receptacles shall light.

b. Place circuit breaker CB11 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #12.- (Convenience circuit breaker CB12)

a. Place circuit breaker CB12 in the ON position. Neon lamp DS13 and test lamps in all convenience receptacles shall light.

b. Place circuit breaker CB12 in the OFF position. All lamps mentioned in step a above shall extinguish.

Circuit #13.- (Spare circuit breaker CB13)

a. Place circuit breaker CB13 in the ON position. Neon lamp DS14 shall light.

b. Place circuit breaker CB13 in the OFF position. Neon lamp DS14 shall extinguish.

Circuit #14.- (CS air conditioner circuit breaker CB14)

a. Place circuit breaker CB14 in the ON position. Neon lamp DS15 shall light.

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b. Place a voltmeter across A and B, A and D, and B and D respectively of the curbside air conditioner mating power connector. Voltage readings of approximately 230 VAC, 115 VAC, and 115 VAC respectively shall be indicated on the voltmeter.

c. Place CB14 in the OFF position. Neon lamp DS15 shall extinguish. The voltmeter shall indicate 0 volts for the same meter connections of step b above.

Circuit #15.- (RS air conditioner circuit breaker CB15)

a. Place circuit breaker CB15 in the ON position. Neon lamp DS16 shall light.

b. Place a voltmeter across A and B, A and D, and B and D respectively of the roadside air conditioner mating power connector. Voltage readings of approximately 230 VAC, 115 VAC and 115 VAC respectively shall be indicated on the voltmeter.

c. Place CB15 in the OFF position. Neon lamp DS15 shall extinguish. The voltmeter shall indicate 0 volts for the same meter connections of step b above.

Circuit #16.- (Main circuit breaker CB16)

a. Place all circuit breakers in the ON position. Neon lamps DS1 thru DS16 shall light. Test lamps in all power receptacle shall light. All fluorescent and incandescent lights shall be ON.

b. Place main circuit breaker CB16 in the OFF position. All lamps, with the exception of the power indicator lamps, shall extinguish.

4.7.3 Signal wiring and components.- (Refer to signal schematic drawings).

4.7.3.1 Continuity resistance of conductors.- Perform the following continuity resistance tests using a calibrated low resistance reading ohmmeter or equivalent. (NOTE: The use of an automatic tester may be substituted to expedite the completion of the following tests. The automatic tester and subsequent test setup must be acceptable by the government technical representative. All test equipment must be calibrated).

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a. Tests of normal thru lines (signal connectors 1 thru 36 to 37 thru 72 respectively).-

(1) Place tester across conductors 1A of signal connectors 1 and 37. The continuity requirements of 3.3.3.1a shall be met.

(2) Repeat step a(1) above for all remaining normal thru conductors (thru 24b) of the same connectors.

(3) Repeat steps a(1) and a(2) above for all remaining normal thru signal connectors 2 thru 36 to 38 thru 72 respectively.

b. Idle line resistance check (signal connectors 1 thru 36 to 37 thru 72 respectively).-

(1) Insert 3 kilo-ohms termination plugs on all Signal 1 (SWBD) Test and Rapid Patch jacks.

(2) Place tester across each pair (A and B) 1 thru 24 respectively of signal connector 1 (SWBD). All resistance readings shall meet the requirements of 3.3.3.1b.

(3) Place tester across each pair (A and B) 1 thru 24 respectively of signal connector 37 (MUX). All resistance readings shall meet the requirements of 3.3.3.1c.

(4) Repeat steps b(1) thru b(3) above with the same 3 kilo-ohms termination plugs inserted on all signal 1 MDF jacks.

(5) Repeat steps b(1) thru b(4) above reversing process, first inserting termination plugs at signal 37 Test and Rapid Patch jacks, and then across signal 37 MDF jacks. All signal 37 (MUX) and signal 1 (SWBD) resistance readings shall meet the requirements of 3.3.3.1b and 3.3.3.1c respectively.

(6) Remove all 3 kilo-ohms termination plugs at the MDF and Rapid Patch panel.

(7) Insert 3 kilo-ohms termination plugs at signals 1 and 37 monitor jacks.

(8) Place tester across each pair (A and B) 1 thru 24 respectively of signal connector 1. The tester shall read 1.5 kilo-ohms $\pm$ 10%.

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(9) Repeat readings of step b(8) above at signal 37.

(10) Repeat steps b(1) thru b(9) above for SWBD signal connectors 2 thru 36 to MUX signal connectors 38 thru 72 respectively.

c. Test of binding posts/signal connectors 79 and 80.-

(1) Place signal 79 CABLE/BINDING POST channel switches in the BINDING POST position.

(2) Insert 3 kilo-ohms termination plugs on all signal 79 MDF jacks.

(3) Place tester across each signal connector 79 pair (A and B) thru pair 24. The tester shall indicate an open circuit on all pairs.

(4) Replace CABLE/BINDING POST channel switches to the CABLE position and repeat test points of step c(3) above. The line termination resistance readings shall meet the requirements of 3.3.3.1b.

(5) Transfer 3 kilo-ohms termination plugs to signal 79 at the Test and Rapid Patch panel and repeat test points of step c(3) above. The line termination resistance readings shall meet the requirements of 3.3.3.1b.

(6) Without removing termination plugs from the Test and Rapid Patch panel, insert additional 3 kilo-ohms termination plugs on all signal 79 monitor jacks.

(7) Repeat test points of step c(3) above. All readings shall indicate 1.5 kilo-ohms $\pm$ 10%.

(8) Remove termination plugs from the monitor jacks.

(9) Replace CABLE/BINDING POST channel switches to the BINDING POST position.

(10) Place tester across each signal 79 binding post pair (A and B) thru pair 24. The line termination resistance readings shall meet the requirements of 3.3.3.1b.

(11) Repeat steps c(1) thru c(10) above for signal 80.

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d. Test of remote alarm and O.W. signals 73 thru 78.-

(1) Place tester between pin 1B of signal connector 73 and pin 15 of remote O.W. signal connector 73 (located behind remote and order wire console). The continuity requirements of 3.3.3.1a shall be met.

(2) Repeat step d(1) above for the following pin terminations: 1A to shield, 2A to 30, 2B to 16, 3A to 17, 3B to 33, 4A to 50, 4B to 4, 5A to 21, 5B to 38, 6A to 36, 6B to 20, 7A to 2, 7B to 19, 8A to 34, 9A to 42, 9B to 46, 10A to 48, 10B to 14, 11A to 25, 11B to 31, 12A to 32, 12B to 39, 13A to 49, 13B to 22, 14A to 5, 14B to 37, 15A to 35, 15B to 3, 16A to 28, 16B to 18, 17A to 6, 17B to 1, 18A to 45, 18B to 47, 19A to 10, 19B to 7, 20A to 44, 20B to 27, 21A to 41, 21B to 12, 22A to 13, 22B to 8, 23A to 29, 23B to 24, 24A to 23, 24B to 26, 25A to 43, 25B to 40, 26A to 11 and 26B to 9.

(3) Repeat steps d(1) and d(2) above for signals 74 thru 78 respectively.

e. Test of pair 25 (phone line).-

(1) Place a shorted plug on phone jack #1 at the MDF 1 patch panel. Continuity to meet the requirements of 3.3.3.1a shall exist across pair 25 of signal connectors 1 thru 8.

(2) Repeat step e(1) above placing shorted plug on indicated phone jack and testing on associated signal connectors: Phone jack #2 to signals 9 thru 16, phone jack #3 to signals 17 thru 20, phone jack #4 to signals 21 thru 24, phone jack #5 to signals 25 thru 28, phone jack #6 to signals 29 thru 32, phone jack #7 to signals 33 and 34, phone jack #8 to signals 35 and 36; and phone jacks 9 thru 16 to signals 37 thru 72 respectively as above.

(3) Place shorted plug on TA-312/PT phone jack at MDF patch bay.

(4) Place tester across each TA-312/PT phone terminal (console and power entrance box). Continuity resistance readings shall meet the requirements of 3.3.3.1a.

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f. Test of pair 26 (intercom line).-

(1) Place shorted plug on intercom jack at MDF patch bay.

(2) Place tester across pair 26 of signal connectors 1 thru 80, and across each intercom terminal. Continuity resistance readings shall meet the requirements of 3.3.3.1a.

g. Test of conference and series circuits.-

(1) Place open plug on one conference jack at the MDF patch bay.

(2) On the associated conference jacks, insert three 3 kilo-ohms termination plugs.

(3) Place tester across the terminals of the open plug. Resistance reading shall be 1 kilo-ohms ± 10%.

(4) Repeat steps g(1) thru g(3) for the remaining sets of conference jacks.

(5) Insert open plug on one phone series jack at the MDF patch bay.

(6) Place tester across the terminals of the open plug. The tester shall indicate 0 ohms.

(7) Alternately place a 3 kilo-ohms termination plug on each phone series jack. The tester shall indicate resistance readings of 3 kilo-ohms ± 10%.

h. Test of SB-22/PT, hybrids and console TH-22/PT circuits.-

(1) Insert 3 kilo-ohms termination plugs on all SB-22 Test and Rapid Patch jacks.

(2) Place tester across each SB-22 pair (located at console) thru pair 24. Resistance readings shall meet the requirements of 3.3.3.1b.

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(3) Place tester across each of the three hybrid pairs (at SB-22). Resistance readings shall be 600 ohms.

(4) Place tester across each of the two TH-22/PT pairs (at SB-22). Resistance readings shall indicate open circuit.

(5) Short the TH-22/PT receive and transmit pairs at the TH-22/PT location. Resistance readings of step h(4) above shall indicate 0 ohms.

(6) Insert open plugs on all hybrid jacks.

(7) Place tester across each open plug terminals. Resistance readings shall be 600 ohms.

(8) Insert shorted plugs on each TH-22/PT jack at the DC patch panel.

(9) Place tester across each DC pair at the console TH-22/PT location. Resistance readings shall indicate 0 ohms.

i. Test equipment circuits continuity check.-

(1) Insert 3 kilo-ohms termination plugs on the following Rapid Patch and TTY patch panel: Telephone TA-341, Oscilloscope AN/USM-296, Transmission Test Set AN/USM-181, and teletypewriters.

(2) At the associated equipment end, place tester across each circuit pair. Resistance readings shall meet the requirements of 3.3.3.1b.

j. Attenuator and amplifiers circuits continuity check.-

(1) Insert an open plug at the first MDF attenuator jack. Insert another open plug at the corresponding monitor jack.

(2) Connect tester first across the tip of the two jacks and than across the ring. The continuity requirements of 3.3.3.1a shall be met.

(3) Insert another open plug at the corresponding Test/ Rapid Patch jack.

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(4) Repeat test of j(2) above. No continuity shall exist.

(5) Repeat test of j(2) above with the tester connected between the open plugs at the monitor and Test/Rapid Patch panels.

(6) Repeat steps j(1) thru j(5) above for all remaining attenuator and amplifier jacks.

4.7.3.2 Dielectric strength and insulation resistance tests.- Perform the following dielectric and insulation resistance tests using a calibrated non-destructive dielectric/insulation resistance test instrument. (NOTE: The use of an automatic tester may be substituted to expedite the completion of the following tests. The automatic tester and subsequent test setup must be acceptable by the government technical representative. All test equipment must be calibrated.)

a. Test of signals 1 thru 72.-

(1) Using appropriate test jigs or test fixtures connect all contacts of signal connector 37 to shelter facility ground terminal.

(2) Connect tester across contact 1A of signal connector 1 and shelter facility ground terminal. Remove corresponding ground connection from contact 1A of signal connector 37.

(3) Adjust tester for a potential of 500 volts DC output and maintain this potential for a continuous period of not less than five seconds. The dielectric strength and insulation resistance requirements of 3.3.3.2 shall be met. Re-establish ground connection to the circuit under test.

(4) Repeat steps a(2) and a(3) above for each of the remaining contacts of the same signal connector; each time ungrounding the contact correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test.

(5) Repeat steps a(1) thru a(4) above for all remaining "normal thru" signal connectors. (NOTE: All testing is accomplished at signal connectors 2 thru 36; grounding at signal connectors 38 thru 72 respectively.)

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b. Test of signals 79 and 80.-

(1) Patch from all MDF signal 79 jacks to corresponding MDF signal 80 jacks.

(2) Place all CABLE/BINDING POST channel switches in the CABLE position.

(3) Repeat steps a(1) thru a(4) above, grounding at signal 80 and testing at signal 79.

(4) Place all CABLE/BINDING POST channel switches in the BINDING POST position.

(5) Repeat steps a(1) thru a(4) above, grounding at binding posts of signal 80 and testing at binding posts of signal 79.

c. Test of signals 73 thru 78.-

(1) Using appropriate test jigs or test fixtures connect all contacts of signal 73 to shelter facility ground terminal.

(2) Remove ground from contact 1A of signal 73. Place tester between contact 1A of signal 73 and shelter facility ground terminal.

(3) Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than five seconds. The dielectric strength and insulation resistance requirements of 3.3.3.2 shall be met. Re-establish ground connection to the circuit under test.

(4) Repeat steps c(2) and c(3) above for each of the remaining contacts of the same signal connector; each time un-grounding the contact correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test.

(5) Repeat steps c(1) thru c(4) for signals 74 thru 78 respectively.

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d. Test of miscellaneous circuits.-

(1) At the MDF patch panel patch between the following jacks:

Signal 1 and SB-22 jacks
Signal 2 and amplifiers 1 - 12 jacks
Signal 3 and amplifiers 13 - 24 jacks
Signal 4 and phone jacks
Signal 5 and conference multiple jacks, intercom jack and all test equipment jacks.

(2) Remove amplifiers from their respective sockets.

(3) Using appropriate test jigs or test fixtures connect all contacts of signal 1 to shelter facility ground terminal.

(4) Remove ground from contact 1A of signal 1. Place tester between contact 1A of signal 1 and shelter facility ground terminal.

(5) Adjust tester to 500 volts DC output and maintain this potential for a continuous period of not less than five seconds. The dielectric strength and insulation resistance requirements of 3.3.3.2 shall be met. Re-establish ground connection to the circuit under test.

(6) Repeat steps d(4) and d(5) above for each of the remaining contacts of the same signal connector; each time ungrounding the contact correlating with the circuit being tested and reconnecting to shelter facility ground following completion of test.

(7) Repeat steps d(3) thru d(6) for signals 2 thru 5 respectively.

4.7.4 Engineering transmission tests.- (See 3.3.4) (NOTE: Test setups other than those indicated must be acceptable by the government technical representative. All test equipment must be calibrated.)

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4.7.4.1 Insertion loss/frequency response.- Insertion loss and frequency response tests to meet the requirements of 3.3.4.1 shall be conducted on 25% of all signal lines in accordance with the following paragraphs:

a. To the SWBD side 4WT pair of the line under test connect a wide band oscillator as shown in figure 1.

b. To the corresponding MUX side 4WT pair of the same line under test connect a wave analyzer across a 600 ohm, 1/4 watt carbon resistor.

c. Terminate input and output adjacent lines with 600 ohms, 1/4 watt carbon resistors.

d. Adjust oscillator to .78 volts (0 db) output at 1 Khz.

e. Adjust analyzer to the exact frequency as the oscillator.

f. Tabulate and record insertion loss in db directly from the analyzer. The requirements of 3.3.4.1a shall be met.

g. Repeat tests of steps d thru e above, of the same line, at the following frequencies: .3, .5, .7, 1.5, 2.0, 2.5, 3.0, 3.5, 16.0, 32.0 and 108 Khz. Tabulate and record the frequency response in db directly from the analyzer at each specified frequency. The requirements of 3.3.4.1b shall be met.

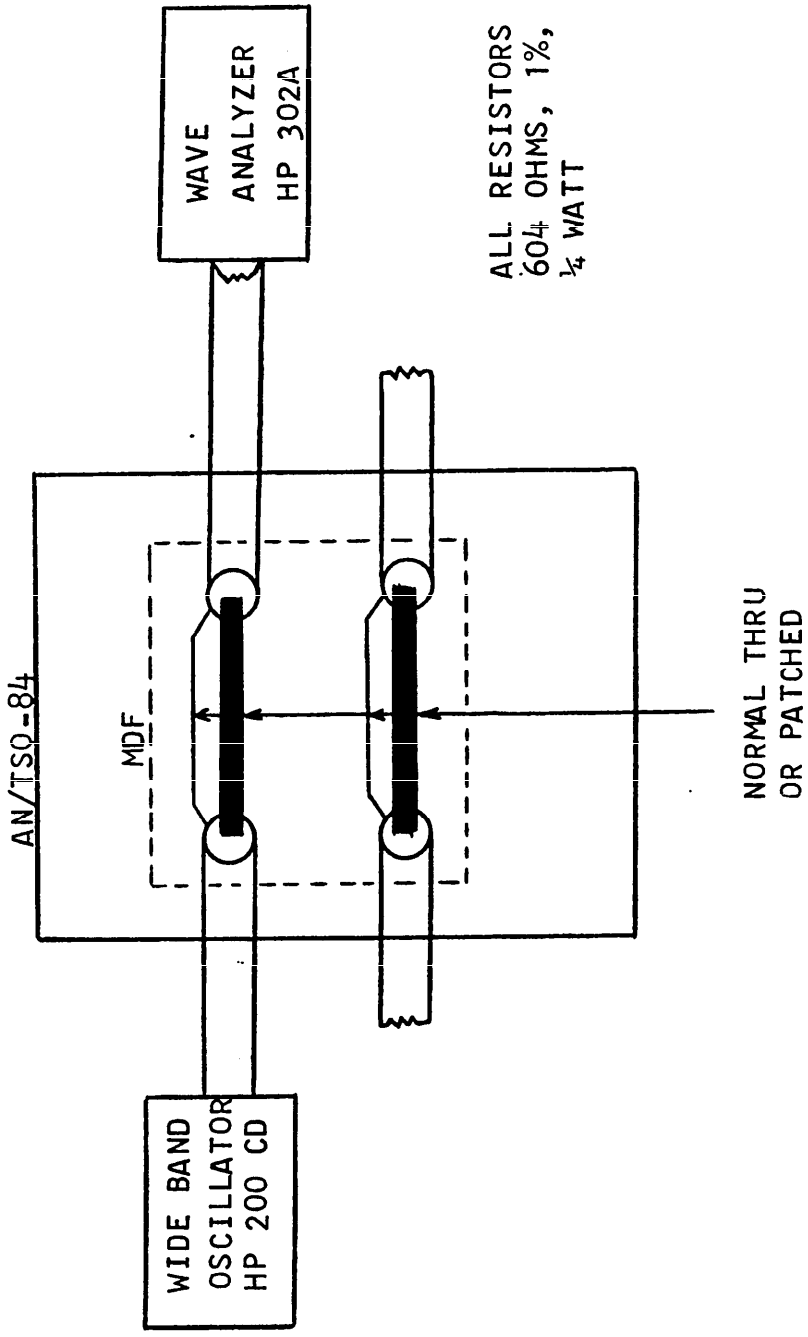
h. Repeat tests of steps a thru g above for 25% of all signal lines. Half of the lines tested shall be normal thru lines; the other half shall be patched at the MDF panels.

(NOTE: Always connect the oscillator to the 4WT pairs when testing at the SWBD side signal connectors; and to the 4WR pairs when testing at the MUX side signal connectors.)

4.7.4.2 Crosstalk.- Perform the following crosstalk tests using the indicated test equipment or equivalent.

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FIG. 1 INSERTION LOSS AND FREQUENCY RESPONSE TEST



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a. To the SWBD side 4WT pair of the line under test connect a wide band oscillator across a 600:600 ohm line transformer as shown in figure 2.

b. To the MUX side 4WR pair of the adjacent channel connect a wave analyzer across a 600:600 ohm line transformer.

c. Terminate all remaining pairs of the same channels with 600 ohms, 1/4 watt carbon resistors.

d. Adjust oscillator to .78 volts (0 db) output at 0.3 Khz.

e. Adjust analyzer to the exact frequency as the oscillator.

f. Tabulate and record crosstalk in db directly from the analyzer. The requirements of 3.3.4.2 shall be met.

g. Repeat tests of steps d thru f above, of the same line, at the following frequencies: .5, .7, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 16.0 and 32.0 Khz.

h. Repeat tests of steps c thru g above with the oscillator connected to the MUX side 4WR pair of the same channel, and the analyzer connected to the SWBD side 4WT pair of the adjacent channel across a 600 ohm, 1/4 watt carbon resistor.

i. Repeat tests of steps a thru h above for 25% of all remaining VF lines. Half of the lines tested shall be normal thru lines; the other half shall be patched at the MDF panels.

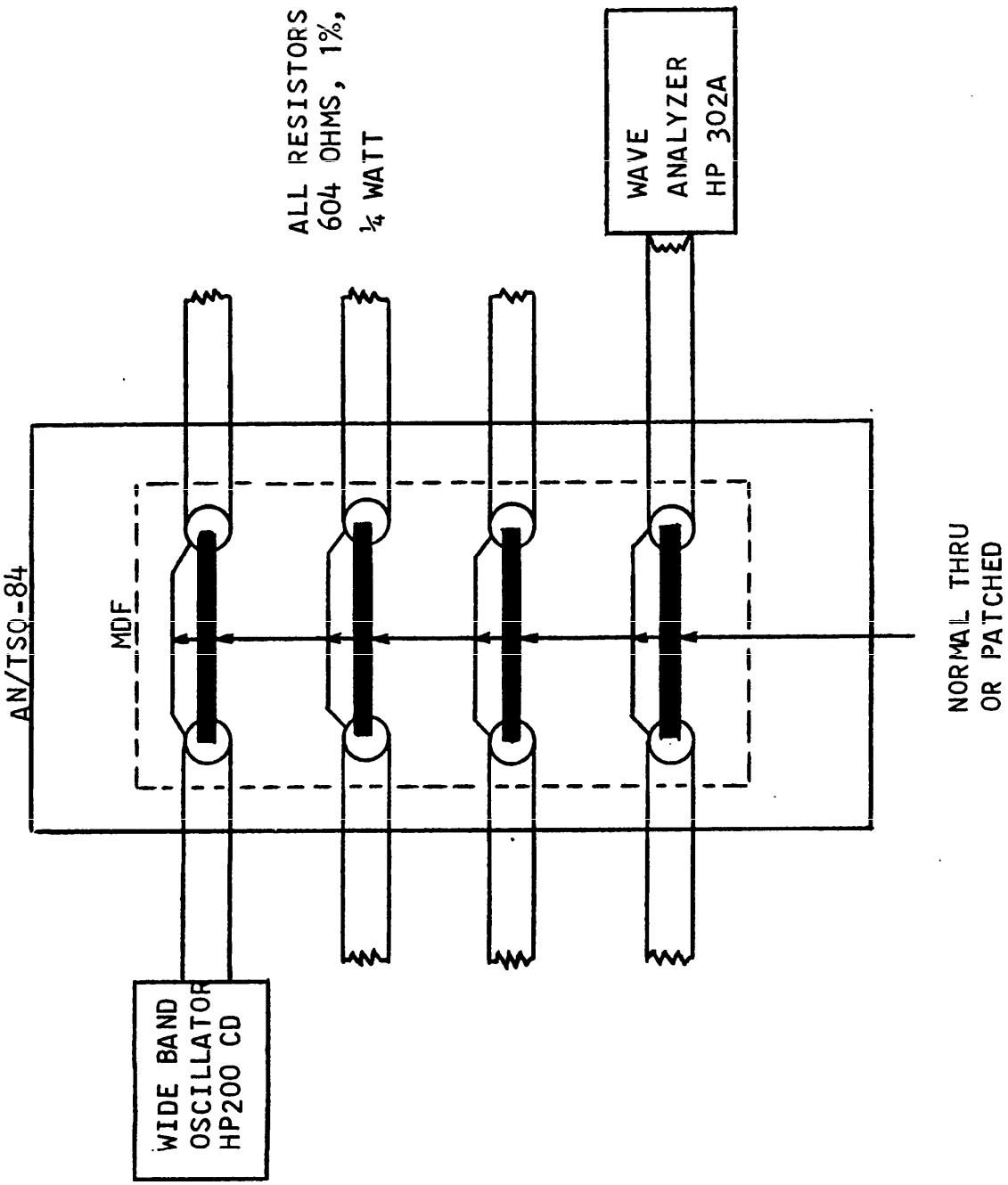
4.7.4.3 Longitudinal balance.- Longitudinal balance tests to meet the requirements of 3.3.4.3 shall be conducted on 25% of all signal lines in accordance with the following paragraphs:

a. To the SWBD side 4WT pair of the line under test connect a wide band oscillator in series with two 300 ohms, 1%, 1/4 watt carbon resistors as shown in figure 3a.

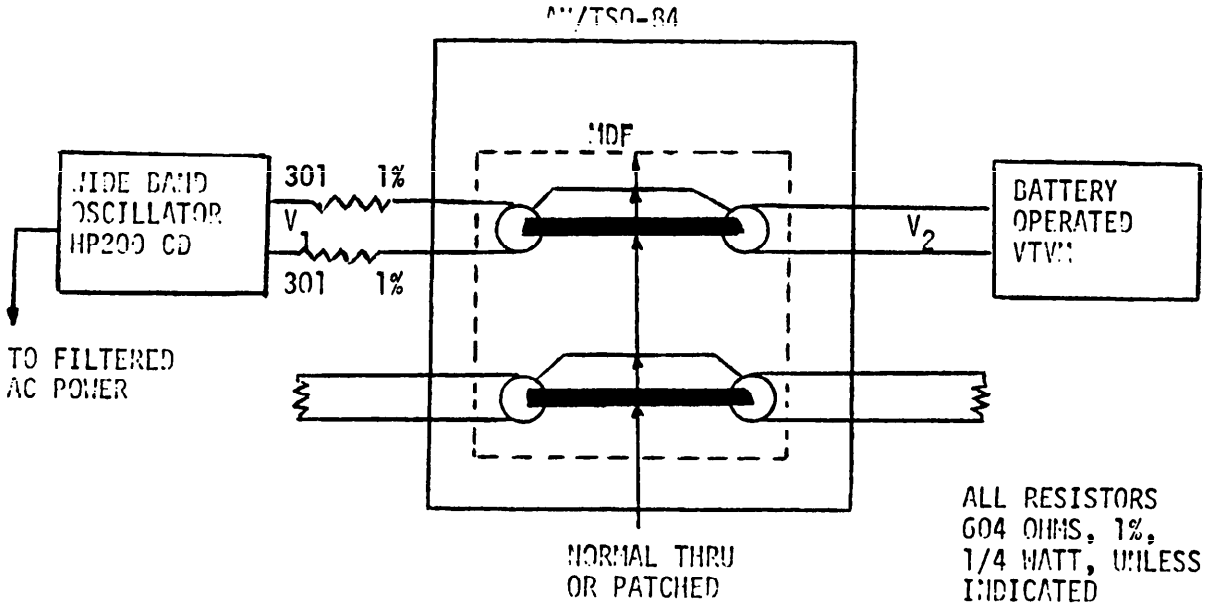
b. To the corresponding MUX side 4WT pair of the same line under test connect a battery operated VTVM across a 600 ohm, 1%, 1/4 watt carbon resistor.

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FIG. 2 CROSSTALK TEST SETUP



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LONGITUDINAL BALANCE TEST SETUP

FIGURE 3a

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c. Terminate input/output adjacent lines with 600 ohms, 1/4 watt carbon resistors.

d. Adjust oscillator to .78 volts (0 db) output at 1.0 Khz.

e. Read and tabulate voltage V2 as indicated in the VTVM.

f. Reconnect the oscillator as shown in figure 3b.

g. Keep output voltage V1 from the oscillator at .78 volts at the same test frequency.

h. Read and tabulate voltage V3 as indicated in the VTVM.

i. Calculate longitudinal balance in db as 20 log V3/V2. The requirements of 3.3.4.3 shall be met.

j. Repeat tests of steps a thru i above for 25% of the lines. Half of the lines tested shall be normal thru lines; the other half shall be patched at the MDF panels.

4.7.4.4 Impedance.- Impedance tests to meet the requirements of 3.3.4.4 shall be conducted on 10% of all signal lines in accordance with the following paragraphs:

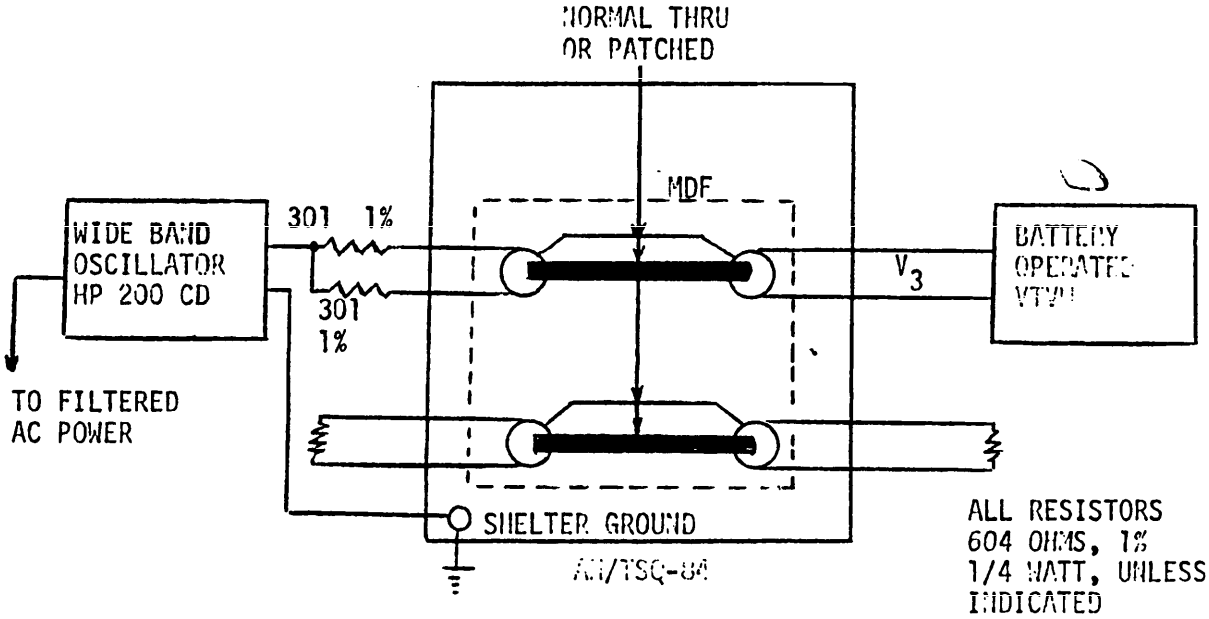
a. To the SWBD side 4WT pair of the line under test connect an impedance bridge meter as shown in figure 4.

b. Terminate corresponding MUX side 4WT pair and input/output adjacent lines with 604 ohms $\pm$ 1%, 1/4 watt carbon resistors.

c. Balance the meter to obtain and record resistances and reactance (inductance or capacitance).

d. Calculate absolute value of impedance as $/Z/ = \sqrt{R^2 + X^2}$, where R is the resistance and X is the reactance. The requirements of 3.3.4.4 shall be met.

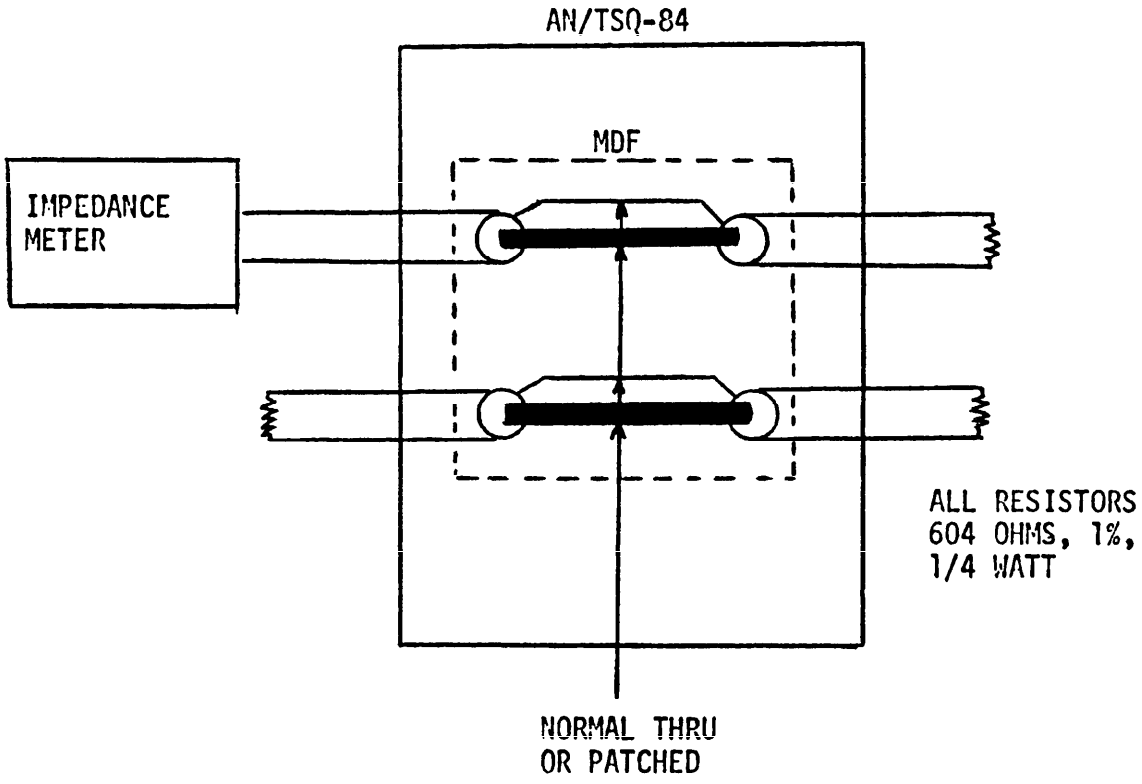
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LONGITUDINAL BALANCE TEST SETUP

FIGURE 3b

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IMPEDANCE TEST SETUP

FIGURE 4

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e. Calculate angle "a" as a = arc tan X/R.

f. Repeat tests of a thru e above with the impedance bridge connected across the MUX side 4WR pair of the same channel and the SWBD side 4WR pair terminated with a 604 ohms ± 1%, 1/4 watt carbon resistor.

g. Repeat tests of a thru f above for 10% of all remaining lines. Half of the lines tested shall be normal thru; the other half shall be patched at the MDF panels.

4.7.5 Test of line conditioning components.-

4.7.5.1 Line amplifiers tests.-

a. Gain and output power.-

(1) Place line amplifier's power supply on.

(2) Place amplifier #1 ON/OFF switch at the MDF patch panel ON. Power indicator lamp for amplifier #1 shall light.

(3) Set audio oscillator (p/o AN/USM-181) to 1,000 hz and adjust output for an indication of -19 dbm on the voltmeter (p/o AN/USM-181). (NOTE: Operate AN/USM-181 in the battery mode for all tests.)

(4) At the test and rapid patch bay patch between the oscillator and input jack of amplifier #1, and between the voltmeter and output jack of amplifier #1.

(5) Adjust amplifier #1 GAIN control until a -19 dbm indication is obtained in the voltmeter. This varifies that the amplifier can be varied down to 0 db.

(6) Adjust same amplifier's GAIN control clockwise until a maximum indication is obtained in the voltmeter. The gain and output power requirements of 3.3.5.1 shall be met.

(7) Repeat steps a(2) thru a(6) above for all remaining amplifiers.

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b. Distortion.-

(1) Retain amplifier #1 ON.

(2) Retain oscillator for a -19 dbm output at 1,000 hz.

(3) Retain amplifier #1 for maximum power output.

(4) Replace voltmeter with a distortion analyzer. Retain or re-establish patch between the distortion analyzer and amplifier #1 output jack.

(5) Adjust distortion analyzer to the exact frequency as the oscillator.

(6) Record distortion reading in percentage directly from the analyzer. The requirements of 3.3.5.1b shall be met.

(7) Repeat steps b(5) and b(6) above with the audio oscillator set to 300, 600, 2,000, and 3,400 hz respectively.

(8) Repeat steps b(1) thru b(7) for all remaining amplifiers.

c. Frequency response.-

(1) Retain amplifier #1 ON.

(2) Retain oscillator for a -19 dbm output at 1,000 hz.

(3) Re-establish voltmeter connection to the output jack of amplifier #1.

(4) Retain amplifier #1 for maximum power output.

(5) Set oscillator frequency to 300 hz and turn dial slowly from 300 to 3,400 hz. The output levels shall not change more than the requirements of 3.3.5.1c.

(6) Repeat steps c(1) thru c(5) above for all remaining amplifiers.

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d. Noise.-

(1) Retain amplifier #1 ON.

(2) Retain oscillator for a -19 dbm output at 1,000 hz.

(3) Retain amplifier #1 for maximum power output.

(4) Remove oscillator's patch at the amplifier's input jack.

(5) Connect a 600 ohm ± 10% terminating resistor to the input jack of the same amplifier.

(6) Replace voltmeter with a noise analyzer.

(7) Record noise readings. The requirements of 3.3.5.1d shall be met.

(8) Repeat steps d(1) thru d(7) above for all remaining amplifiers.

4.7.5.2 Attenuators tests.-

a. Patch audio oscillator (p/o AN/USM-181) to attenuator #1 input jack at the test and rapid patch bay.

b. Patch voltmeter (p/o AN/USM-181) to attenuator #1 input jack at the monitor patch bay.

c. Set oscillator to 1,000 hz and adjust output for an indication of 0 dbm on the voltmeter.

d. Remove voltmeter's patch from the attenuator's input jack and connect to the output jack of the same attenuator.

e. Rotate attenuator's switch fully clockwise.

f. Record attenuation reading in db directly from the voltmeter. The requirements of 3.3.5.2a shall be met.

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g. Rotate attenuator's switch counterclockwise, one step at a time, each time recording the attenuation reading on the voltmeter. The requirements of 3.3.5.2b shall be met.

h. Repeat steps a thru g above for all remaining attenuators.

4.7.5.3 Line equalizers tests.- (See 3.3.5.3)

a. Patch audio oscillator (p/o AN/USM-181) to attenuator #1 input jack.

b. Patch output of attenuator #1 to input of line amplifier #1.

c. Patch output of line amplifier #1 to input of line equalizer #1.

d. Patch output of line equalizer #1 to voltmeter (p/o AN/USM-181).

e. Set audio oscillator to 0 db output at 3 Khz; attenuator to 10 db; line amplifier ON and set at minimum amplification; and line equalizer at maximum (maximum clockwise rotation of the control).

f. Vary control of line equalizer from maximum to minimum. Voltmeter should indicate a compensation of approximately 17 db. Reset control to maximum.

g. Adjust gain of line amplifier so that the output of the line equalizer is - 10 dbm.

h. Set audio oscillator to 300 hz retaining 0 db output from the oscillator.

i. Adjust control on the equalizer so that the voltmeter reads - 10 dbm at 300 hz.

j. Repeat procedures of step g thru i until the voltmeter indicates identical readings for 300 hz and 3 Khz. This verifies that the equalizer can be compensated to produce an approximately flat frequency response.

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4.8 Operational tests.- (See 3.4)

4.8.1 Switchboard SB-22/PT lines.-

a. Open the rear cover of the switchboard to access battery case. Insert two Batteries BA-30 into each end of the battery case and replace battery case into the contacts of the SB-22. Connect all switchboard circuits to the appropriate binding posts at the back of the SB-22. Connect Telering TA-248 to the appropriate connection at the rear of the switchboard.

b. Insert two Batteries BA-30 into the console Telephone Set TA-312/PT. Connect phone wires to the TA-312/PT.

c. At the MDF jack panel, patch between the TA-312 jack and the SB-22 #1 jack.

d. Operate visual and audible alarm switch on operator's pack of switchboard to VIS.

e. Ring from the TA-312/PT. The line signal on the line pack shall operate to white and the light on the operator's pack should light.

f. Insert the plug of the operator's cord into the jack of line pack #1. The line signal shall operate to black and the light on the operator's pack shall go out.

g. Operate the switch in the cord of the operator's telephone set (push-to-talk switch) to the first position and communicate between the phones.

h. Remove operator's cord from the line pack and operate the visual and alarm switch from VIS to AUD.

i. Ring from the TA-312/PT. The line signal on the line pack shall operate from black to white and the alarm buzzer shall sound.

j. Insert the plug of the operator's cord into the jack of the line pack; the line signal shall return to black and the buzzer shall stop.

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k. Push the RING BACK-PWR RING FWD switch to the RING BACK position and turn the hand ringing generator on the operator's pack. The TA-312/PT shall ring.

l. Remove the plug of the operator's cord from the jack of the line pack and insert it into the jack of the adjacent line pack. Insert the plug of the cord on the second line pack into the jack of the line pack connected to the TA-312/PT.

m. Turn the hand ringing generator on the operator's pack; the TA-312/PT shall ring.

n. Repeat steps c thru m for SB-22 circuits #2 thru #24 respectively.

o. Remove the plugs from the jacks and insert the plug of the operator's cord into the jack of the operator's pack.

p. Pull out the LITE-OUT-NA-IN knob; the lamp shall light.

q. Push the LITE-OUT-NA-IN knob into its original position; the lamp shall extinguish.

4.8.2 Hybrid lines.-

a. Perform indicated connections as shown in figure 5.

b. Communicate from phone #2 to phone #3. Phone #1 shall not receive the transmitted phone message from phone #2.

c. Communicate from phone #3 to phone #1. Phone #2 shall not receive the transmitted phone message from phone #3.

d. Repeat steps a thru c above for the remaining two hybrids.

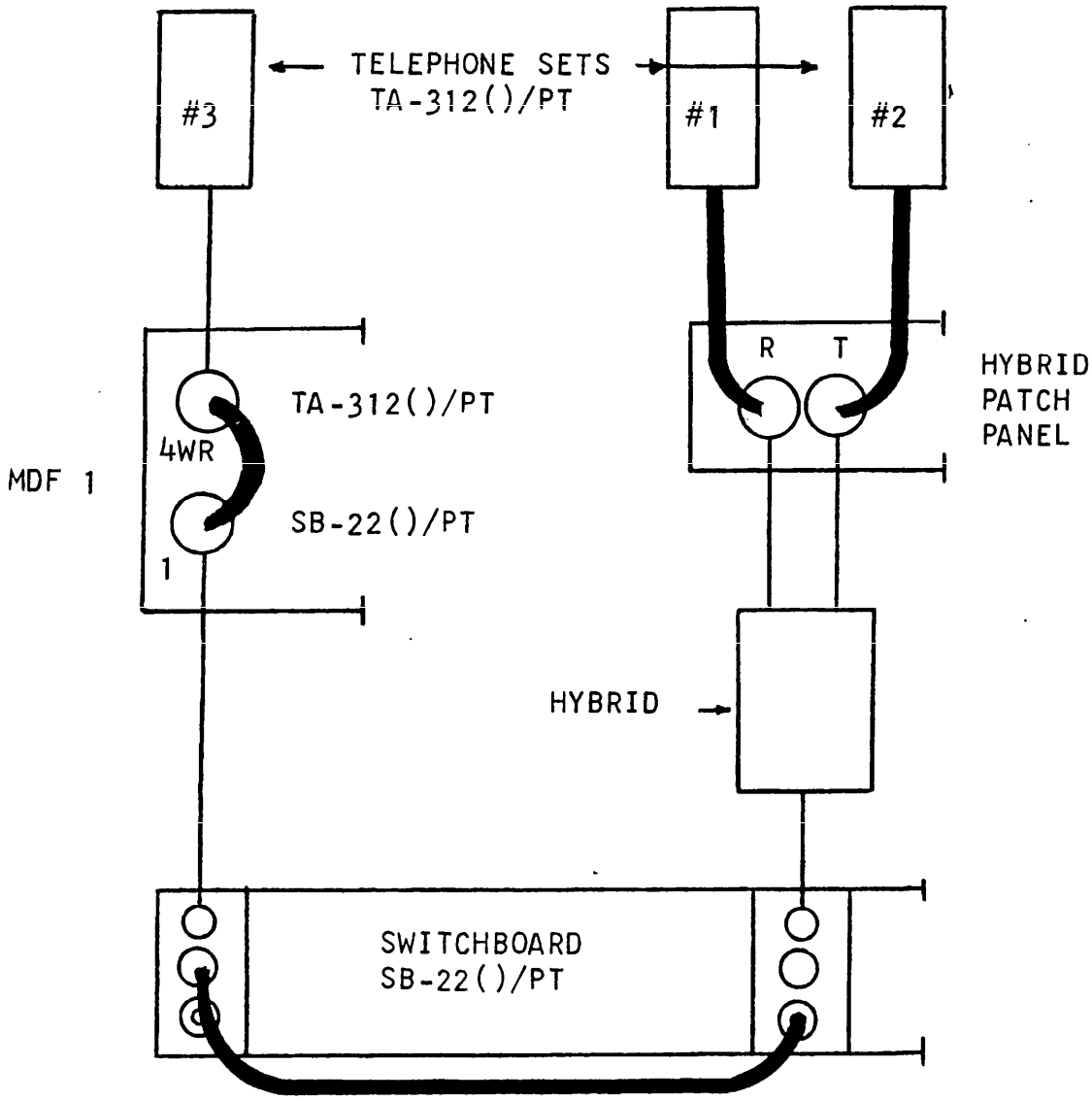
4.8.3 Teletype lines.-

a. Perform patch connections as shown in figure 6.

b. Arrange teletypewriters for full duplex, 20 ma operation.

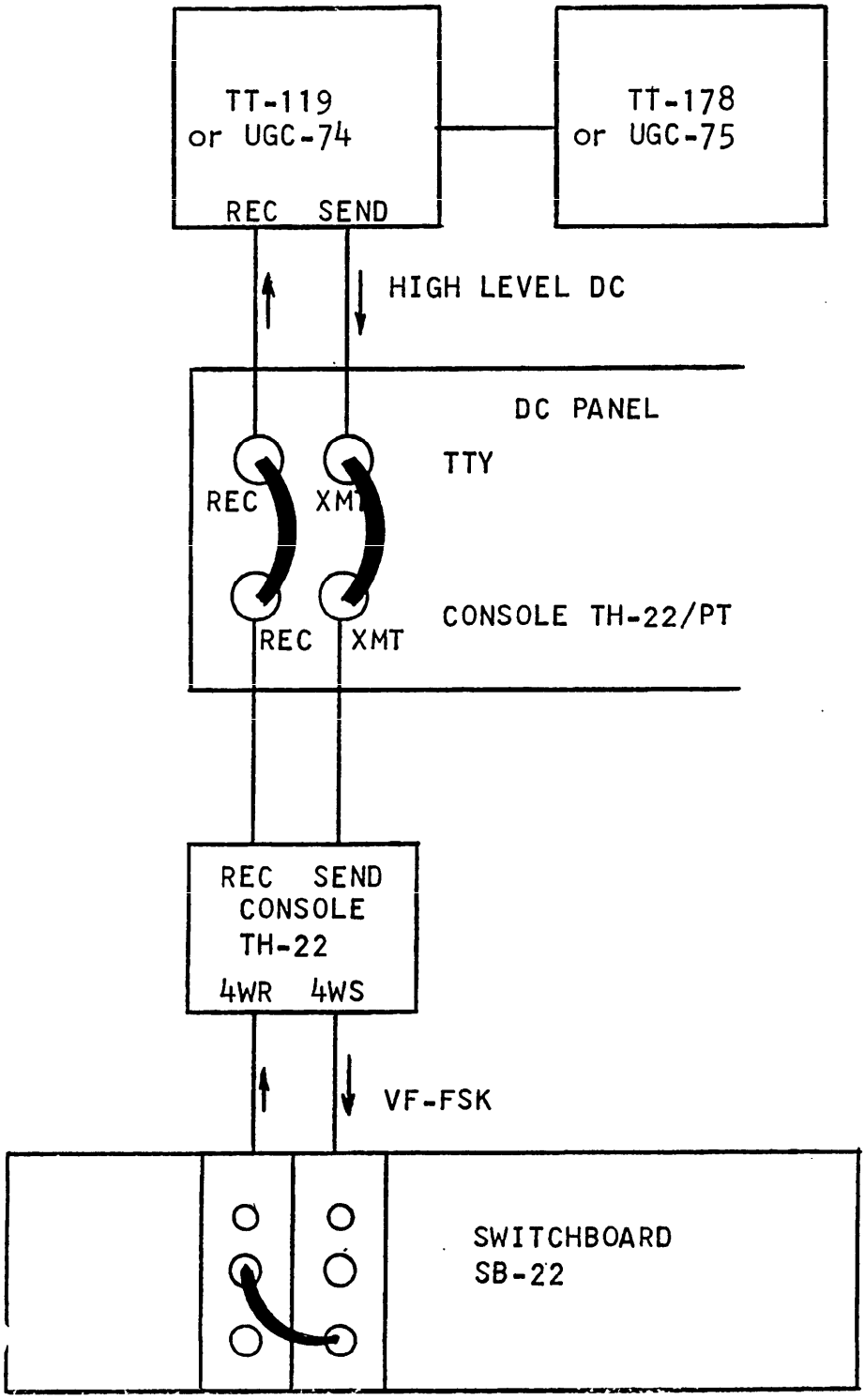
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FIG. 5 HYBRIDS OPERATIONAL CHECK



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FIG. 6 TELETYPEWRITERS OPERATIONAL CHECK



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c. Set NORM/REC/SEND switch on the TH-22/TG to NORM; 4W/2W/TEL switch to 4W MODE ONE/MODE TWO switch to MODE ONE and VF-20 switch to VF. Connect line and loop side of TH-22/TG as shown in figure 6.

d. Send a series of "FOX" transmission on the teletypewriter keyboard. The transmitted message shall be printed on the page printer.

4.8.4 Test of Amplifier-Headset AM-6789/G.-

a. Insert two 9 volt batteries BA-3090 or equivalent into the unit.

b. Plug the power cord into 115 VAC, 50/60 hz power line receptacle.

c. Plug the transmission line connector (U-77/U) into the U-79/U receptacle on the back of the unit.

d. Plug a H-333( )/U Headset-Microphone into the receptacle (U-183/U) on the front of the unit.

e. Set mode switch to 2-wire.

f. Turn power switch to LINE. The white light shall come on.

g. Test for sidetone by talking into headset-microphone. Repeat with mode switch in the 4-wire position.

h. Set power switch to BATTERY. The white light shall extinguish.

i. Repeat sidetone tests of paragraph g above. Reset mode switch to 2-wire and power switch to LINE.

j. Insert proper batteries to Telephone Sets TA-312/PT and TA-341/TT.

k. Patch TA-312/PT into 2WT/R side of AM-6789/G.

l. Communicate between the TA-312/PT and H-333( )/U Headset-Microphone.

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m. Remove patch cable. Reset mode switch of AM-6789/G to 4-wire. Patch between TA-341/TT and AM-6789/G (receive to transmit and transmit to receive respectively.)

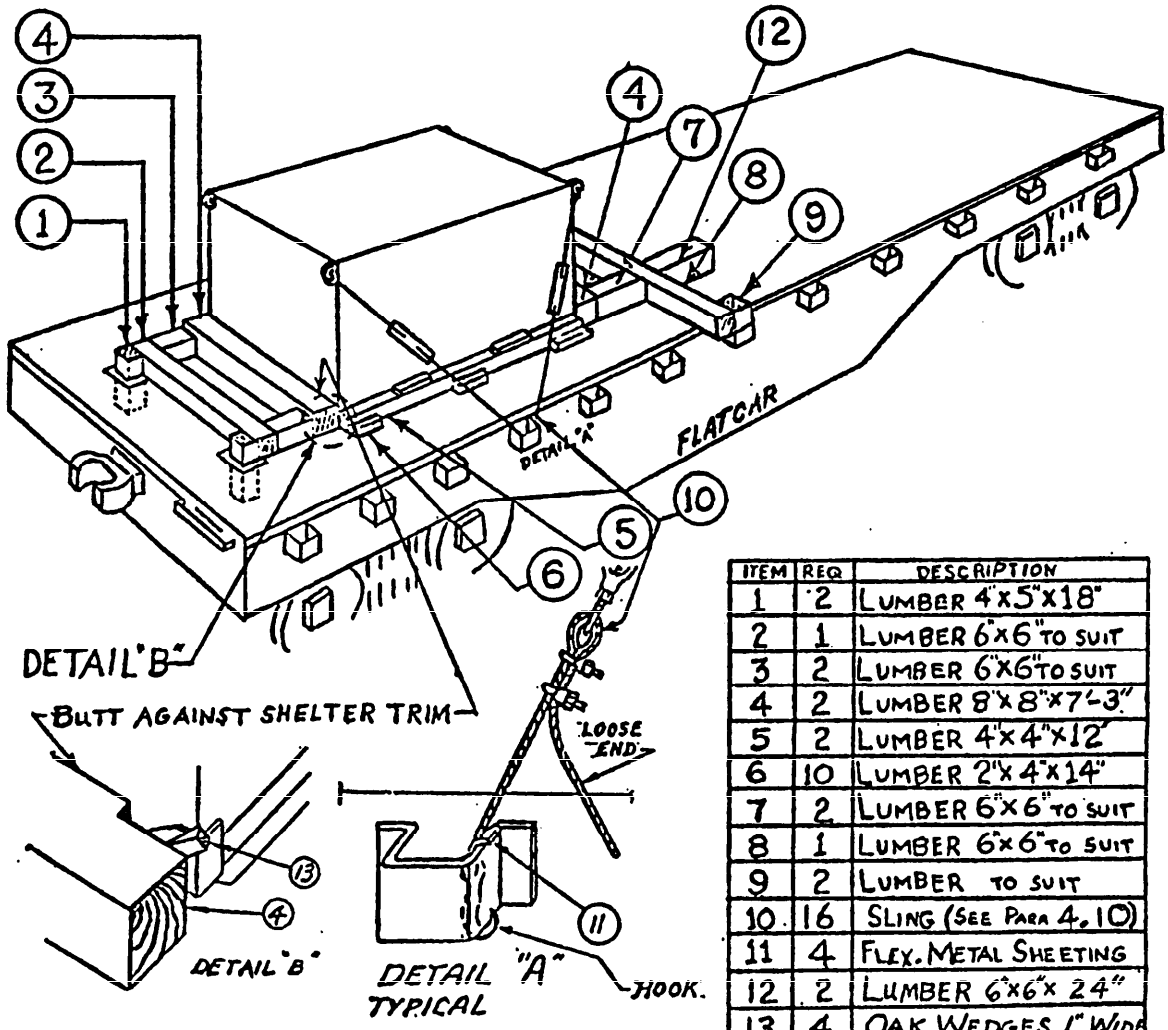
n. Communicate between the TA-341/TT and the H-333( )/U Headset-Microphone.

4.9 Vehicular transportation test.- The facility shall be loaded to simulate the AN/TSQ-84( ) using live and dummy loads (See 6.2b). The air conditioners shall be mounted outside the shelter facility in their normal operating position. Using the lift and tie down slings furnished with the bare shelter, the facility shall be secured as indicated on the shelter transportation data plate, to the bed of a government furnished 2-1/2 ton, M-35 truck. The tire pressure shall be set for cross-country travel as listed on the truck dashboard. The facility shall be centralized and blocked to prevent shifting. The facility shall be transported over the specially prepared course described in 4.9.1 for 10 laps at each speed of 5, 10, 15 and 18-20 MPH. A lap is defined as traversing the course in one direction. Upon completion of this test, the facility shall not have sustained any of the types of damage cited in 3.5.

4.9.1 Vehicular transportation course.- The course for the military truck transportation test shall have twelve (12) 4-inch by 8-inch timbers placed 25 feet apart on the 8-inch face and with the 4-inch face fully above ground. The top edges shall have a 1 x 1-inch chamfer. The sixth and twelfth timbers shall be placed at 45 degrees to the direction of travel, and all other timbers shall be placed perpendicular to the direction of travel. Each timber shall be anchored down securely by any effective means. The ground on which the timbers are placed shall be hard and level.

4.10 Rail transport, impact test.- The shelter facility loaded as specified in paragraph 4.9, secured to a flatcar (the test flatcar) and coupled to two other flatcars shall be impacted by a loaded gondola car. The test shall be performed on a straight and level stretch of track. The test flatcar with its timber blocking and shelter facility tie down slings (contractor furnished), shall be as specified in figure 7 and shall meet the provisions of the test as specified in paragraph 4.10.1. The test flatcar (less the shelter facility) and the two empty flatcars coupled to it shall each weigh 45,000 to 55,000 pounds. The loaded gondola car

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ITEM	REQ	DESCRIPTION
1	2	LUMBER 4"x5"x18"
2	1	LUMBER 6"x6" TO SUIT
3	2	LUMBER 6"x6" TO SUIT
4	2	LUMBER 8"x8"x7'-3"
5	2	LUMBER 4"x4"x12'
6	10	LUMBER 2"x4"x14"
7	2	LUMBER 6"x6" TO SUIT
8	1	LUMBER 6"x6" TO SUIT
9	2	LUMBER TO SUIT
10	16	SLING (SEE PARA 4.10)
11	4	FLEX. METAL SHEETING
12	2	LUMBER 6"x6"x 24"
13	4	OAK WEDGES 1" WIDE

NOTES:
a. The blocking lumber shall be notched to accomodate towing eyes, and bolts. Wedge item 13 between towing corner forging and item 4 notch (Detail B)
b. The blocking lumber shall be rigidly fastened to the floor of the flatcar.

FIGURE 7

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shall meet the provisions of the test as specified in paragraph 4.10.2 shall have a total weight of 165,000 pounds. The loaded gondola car totaling 165,000 pounds, traveling at nine miles per hour (or equivalent energy conditions acceptable to the government technical representative), shall be impacted against the stationary loaded test car coupled to the two empty flatcars, all with brakes off. All cars shall have standard (rigid) draft gear. The test shall consist of four impacts, two from each end of the shelter facility. The lashing, blocking and wedges holding the shelter facility in place shall be tight at the start of each run. If the blocking is torn loose or the lashing is broken during impact, the run shall be repeated. After each impact the shelter facility shall meet the requirements of paragraph 3.6.

4.10.1 Test flatcar.- The wooden planking of the test flatcar bed shall be in good condition. All 6 x 6 inch blocking shall be secured to the bed of the flatcar using 10 inch spikes on 12 inch centers. The blocking along the door end and opposite end of the shelter shall fit tightly against the corner castings along the full length of the castings except as necessary to clear the towing shackles, this shall be accomplished by means of sawed notches. The blocking along the other two sides of the shelter shall fit tightly against the corner castings, along the full length of the castings. These four blocking members shall be notched or shimmed to contact the trim along the four sides of the shelter. The 4 x 4 and 2 x 4 inch blocking shall be nailed to the bed of the flatcar as shown in the figure referred above. All other blocking shall be firmly secured as shown in the figure. Lashing shall consist of four wire rope slings, each including 3/8 inch diameter 6 x 19, flexible steel, preformed wire rope (having a breaking strength of 10,600 pounds minimum), a turnbuckle, two hooks, with thimbles and fasteners as required; the breaking strength of the sling shall be 9,000 pounds minimum. Before each impact, each sling shall be taut (600 - 800 pounds tension).

4.10.2 Loaded gondola car.- The load in the loaded gondola car shall be one of the following:

a. A coarse size of either coal, crushed stone, gravel, crushed concrete or metal scrap (other than sheet metal, metal shavings or other springy forms of metal).

b. Chunk, slab, block or plate forms of the preceding acceptable materials, constrained to prevent shifting.

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4.11 Drop tests.-

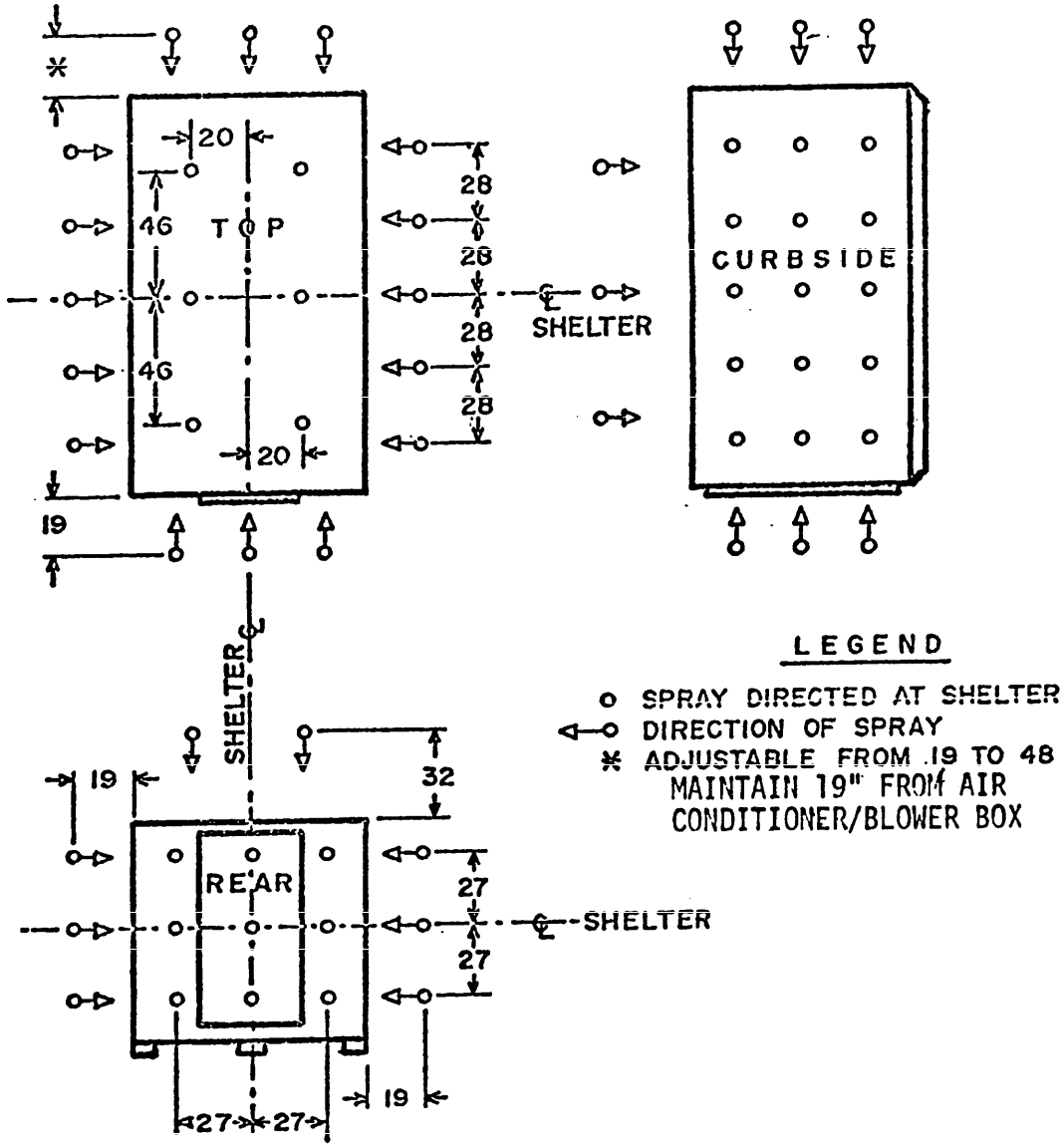
4.11.1 Flat drop.- The shelter facility loaded as in 4.9, shall be lifted 18 inches from the ground, leveled, and allowed to fall freely with the skids impacting onto a hard level concrete surface. The 18 inch measurement shall be taken from the bottom of the skids. The test shall be performed once. Upon completion of this drop, the facility shall meet the requirements of 3.7.

4.11.2 Rotational drop.- The shelter facility loaded as in 4.9 shall have one skid resting on a nominal 4 inch high timber. The outer edge of the opposite skid shall be raised 18 inches and the shelter facility shall be allowed to fall freely onto a hard substantial concrete surface. This impact shall be performed four times, once for each bottom edge of the shelter facility. On completion of these drops, the shelter facility shall meet the requirements of 3.7.

4.12 Watertightness test.- The shelter facility, without the use of any additional external sealing, caulking, taping, etc., shall be subjected to the following two part test: (NOTE: Perform tests with inside cover of entrance boxes removed).

a. Each of the five exposed surfaces of the shelter facility, with covers closed, shall be sprayed with water from nozzles, model G29SQ (or GG29SQ) as made by Spraying Systems Company, Bellwood, Illinois, or equal, pointing directly at the surface and arranged to provide uniform coverage of the surface. Each end panel shall be tested using nine nozzles, each side using fifteen nozzles, and the roof using six nozzles. Each nozzle shall operate at a pressure of forty psig at the nozzle, and shall be located nineteen inches from the panel with the exception of the roof nozzles which shall be at a distance of thirty-two inches. All of the five surfaces shall be simultaneously subjected to the test of this paragraph unless otherwise specified by the contracting officer. (See figure 8 for nozzle location.) Each panel shall be sprayed for a period of 20 minutes. (If the watertightness test is to be followed by the fording test of 4.13, the bottom row of nozzles may be eliminated with the exception of the nozzles facing the door-end panel.) Upon completion of the above test, the shelter facility shall meet the requirements of 3.8.

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NOTES

1 NOZZLES SHALL BE PERPENDICULAR TO SHELTER.
2 UNLESS OTHERWISE SHOWN ALL DIMENSIONS SHALL BE TAKEN FROM INDICATED CENTER LINES.
3 GAGES SHALL BE PERMANENTLY INSTALLED IN PIPING TO INDICATE PRESSURE AT REMOTE NOZZLES.
4 ROADSIDE NOZZLE LOCATIONS SAME AS CURBSIDE
5 FRONT NOZZLE LOCATIONS SAME AS REAR.
6 DIMENSIONS ARE IN INCHES.

NOZZLE LOCATIONS

FIGURE 9
PAGE 57

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(NOTE: For first article and Group C tests only, the air conditioners, with brackets, shall be mounted outside the shelter in the operating position. For Group B tests the air conditioner openings shall be secured with the designated blank off covers.)

b. With the air conditioners and bracketry removed, the openings secured with the air conditioner opening blank off covers, and all other covers opened to their operating position (except the shelter door vent panel) repeat a above.

4.13 Fording test.- The shelter facility shall be immersed in water for one hour to a depth of 21 inches (measured from the bottom of the skids) and meet the requirements of 3.9. No special fording kits shall be used. The shelter shall be restrained from floating by external tie downs. To determine that no water has entered the shelter panels, holes shall be made at points to be determined by the government at the time of testing. After this determination, the holes shall be sealed with rivets and sealer.

4.14 Environmental, shelter assemblage.- (See 3.10 and 6.8)

4.14.1 Temperature tests.- The temperature tests on the assemblage shall be performed in accordance with Methods 501 and 502, Procedure I, of MIL-STD-810 with the following limits:

a. High storage temperature of 160°F for 48 hours.

b. High operating temperature of 125°F.

c. Low storage temperature of -70°F for 24 hours.

d. Low operating temperature of -25°F.

4.14.1.1 Preconditioning.- The assemblage shall be preconditioned prior to the tests for 24 hours at the following conditions with the shelter door open:

a. Temperature 77° ± 5°F

b. Relative Humidity 50 ± 5 percent

c. Local ambient barometric pressure corrected to between 28 and 32 inches of mercury.

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4.14.1.2 High temperature.- Subject the assemblage to Test Method 501, Procedure I of MIL-STD-810 with the following exception: After step 5, with shelter door closed, operate the air conditioners until the air temperature inside the shelter reaches equilibrium. Record time to reach equilibrium and temperature maintained by the air conditioners.

4.14.1.3 Low temperature.- Subject the assemblage to Test Method 502, Procedure I of MIL-STD-810 with the following additions and exceptions:

a. Precondition the shelter assemblage in accordance with 4.14.1.1 above.

b. Steps 2 and 4 shall be performed with the shelter door, louvers and vents closed.

c. Change step 5 as follows: "With the shelter door, louvers and vents closed initiate operation of the air conditioners in the heating mode, and the electric heaters. (NOTE: The electric heaters will cycle on and off during the early stages of warm up until the fan reaches continuous rotation.) After two hours of warm up operate all the equipment in the assemblage except the Teletypewriters and the Line Amplifiers. When the temperature of the air in the Teletypewriter and Line Amplifier rack space reaches 32°F, operate these equipments. Compare the results with the data obtained in accordance with section 3, paragraph 3.2 of MIL-STD-810."

4.14.1.4 Humidity test.- Subject the assemblage to Test Method 507, Procedure II of MIL-STD-810. During steps 2, 3 and 7 the shelter door, louvers and vents shall be open. For the remainder of the test, except for necessary intermittent access of test personnel, the door and louvers shall be closed. Prior to measurements, excess surface water may be removed by wiping external surfaces only. In step 6, measurements shall be taken during the last five hours of the last cycle.

4.15 Humidity, hardware.- A hardware sample of five of each size and type (if the total quantity of any size or type is less than five, all shall be tested) of bolt, screw, nut, flat washers, lock washers and rivnuts shall be randomly selected for test. The hardware sample shall be subjected to the humidity test specified in Method 507, Procedure II of MIL-STD-810 to determine conformance to the requirements of paragraph 3.11.

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4.16 Mounting insert tests.-

4.16.1 First article and Group C insert test.- Four of each size insert shall be randomly selected for test. Each selected insert shall be tested with an axial load followed by a torque load in accordance with Table V. After test, the requirements of 3.12 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc. (Inserts in upper front wall members shall not be subjected to this test.)

4.16.2 Production insert test.- Five of each size insert (if the total quantity of any size is less than five, all of that size shall be tested) shall be randomly selected for test in each sample shelter facility. Each selected insert shall be tested in accordance with Table V. After test, the requirements of 3.12 shall be met. Failure of any insert shall be cause for rejection of the sample shelter facility. This test shall be conducted prior to installation of brackets, racks, etc.

TABLE V - Mounting Inserts Loads

Insert Size	Preproduction Test Axial(lbs)	Preproduction Test Torque(in lbs)	Production Test Axial(lbs)	Production Test Torque(in lbs)
5/16-18	2000	100	1000	100
1/4-20	1300	60	650	60
#10-32	600	23	300	23

4.17 Acoustical noise test.- The shelter facility shall be located in an area remote from excessive machinery, vehicle and other noise generating sources to reduce ambient noise to a minimum, with the shelter door closed. Only necessary test personnel shall be present in the shelter. Using Octave Band Analyzer, General Radio Corp. Model 1558A, or equal, and Microphone, Piezoelectric, General Radio Corp. Model 1560-P6, or equal, record ambient noise levels over the frequency ranges specified in 3.13 at the operator's positions. Operate the air conditioners in the cooling, high fan speed mode, and the operational equipment (except teletypewriters) and repeat noise level recordings in the same frequency levels and positions as above. The noise levels, corrected for ambient shall meet the requirements of 3.13.

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4.17.1 Measurement location.- Noise measurements shall be made at the following locations:

a. Center aisle of shelter, 65" above floor, 42" from front wall.

b. Center aisle of shelter, 65" above floor, 52" from rear wall.

c. Thirty-six (36) inches from curbside wall, 49" above floor, 34" from rear wall.

d. Thirty-six (36) inches from curbside wall, 49" above floor, 16" from front wall.

4.18 Testing of solderless terminal-lug connections.- Connections made with solderless terminal-lugs shall be tested for conformance to 3.22.4.

4.19 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.23. When a dimension is not within specified or design limits, it shall be considered a defect.

a. Size and mounting dimensions of brackets and mounting plates.

b. Size and mounting dimensions of racks and cabinets including equipment mounting inserts.

4.20 Visual and mechanical inspection.- The shelter facility shall be examined for the defects listed in Standard MIL-STD-252, the requirements of 3.24 and Table VI. (NOTE: All welding is to be inspected prior to application of paint finish.)

TABLE VI - Classification of Visual and Mechanical Defects

Classification	Defects
Major	1. Mounting plates missing or mislocated. 2. Aircraft loading data plate, instruction or nameplate omitted, incorrect or illegible.

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TABLE VI - Classification of Visual and Mechanical Defects (contd)

	3.	Welds - cracked or porous.
	4.	Sealer improperly applied (holes, separations, or lack of adhesion).
	5.	Mounting inserts missing or mislocated.
	6.	Burrs or sharp edges in wire duct not removed.
	7.	Equipment or bracketry loose or missing.
	8.	Doors, etc., inoperative.
	9.	Electrical receptacles, switches, jacks, connectors, wire ducts or other electrical components loose, improperly located or inoperative.
	10.	Locking or holding devices missing or inoperative.
	11.	Gaps between sections of wire duct exceed 1/16".
	12.	Schematics missing.
	13.	The air conditioners cover does not align properly with the mounting inserts of the shelter facility.
	14.	Ground stud and other external panel hardware not properly sealed.
	15.	Bulkhead power and signal connector fasteners not sufficiently tight.
Minor	1.	Doors not easily operable.
	2.	Fastening devices difficult to operate.
	3.	Finish - abrasions or scratches
	4.	Burrs and sharp edges on bracket not removed.
	5.	Screws loose.
	6.	Gaskets, not completely and securely adhered.
	7.	Schematics not properly adhered to wall.

4.21 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

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4.22 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling.

Paragraph 4.7.1 thru 4.7.3.2 Power and Signal Wiring

Paragraph 4.20 Visual and Mechanical

4.23 Reliability demonstration.- The requirements of paragraph 3.25.1 shall be demonstrated by use of a data collection system. Failure data shall be collected, as a minimum, during all government acceptance testing on all systems. The following data shall be collected, as a minimum, on each failure or apparent malfunction:

a. Time of failure or malfunction

b. Time to repair

c. Failure or malfunction symptoms

d. Part designation

e. Description of failure

f. Repair (corrective) actions

g. Analysis of failure or malfunction

The total operation hours on each system and the total piece part operating hours for each type of piece part shall be recorded. Paragraphs 5.5 through 5.10 inclusive of MIL-STD-781 shall apply.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation, packaging, and packing of the shelter facility shall be in accordance with MIL-S-55505.

6. NOTES

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6.1 Intended use.- The Shelter, Electrical Equipment S-538( )/TSQ-84 is the shelter facility of the AN/TSQ-84( ). The shelter facility is used to house the following operational equipments forming the Communications Technical Control Center:

a. 2 ea Air Conditioner, Horizontal Compact, 9000 BTU.

b. 1 ea H-156 Hand Set.

c. 1 ea AN/USM-296 Oscilloscope.

d. 1 ea TH-22/TG Terminal, Telegraph.

e. 1 ea AN/USM-181 Test Set, Telephone.

f. 1 ea AN/GGM-15(V) Test Set, Telegraph.

*g. 1 ea TT-178/FG Reperforator Transmitter.

*h. 1 ea TT-119( )/FG Teletypewriter.

i. 2 ea SB-22/PT Switchboard.

j. 1 ea TA-312/PT Telephone.

k. 1 ea TA-341/TT Telephone.

l. 1 ea Amplifier Interface, Test Head Set, AM-6789( )/TSQ-84.

m. 1 ea Head Set, Microphone, H-333( )/U.

n. 1 ea Power Supply, 24 VDC, PP-7271/U.

o. 1 ea Power Supply, PP-6145/TSC-62.

p. 18 ea Amplifiers, AM-6120/TSC-62.

q. 2 ea Head Set, H-16/U.

r. 9 ea Equalizers, Telephone Line, TA-957/G.

s. 2 ea Intercommunication Station, LS-147( )/FI.

t. 1 ea Telering, TA-248/TT.

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* NOTE: The shelter facility provides mounting and interface for substitution of AN/UGC-74( ) Teletypewriter and AN/UGC-75( ) Teletypewriter for these equipments.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. First Article and Group C inspection of the shelter facility requires use of contractor fabricated dummy loads or the government furnished live loads (operational equipments) as specified in the bid request and contract.

c. A list of government furnished equipments required as part of the S-538( )/TSQ-84.

d. A list of equipments to be government loaned for testing purposes.

6.3 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.- Examination consists of simple generally non-destructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.- Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.- Approval to ship may be withheld, at the discretion of the government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.5)

6.5 Nomenclature.- The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: S-538R/TSQ-84. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

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6.6 Verification inspection.- Verification by the government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.7 Tropicalization of material.- All materials of construction are specified on the drawings and are fungus-inert. Substitution of any materials other than specified must be approved by the contracting officer. The material will only be approved after the contractor tests the material for fungus-resistance satisfactorily in accordance with requirement 508 of MIL-STD-810.

6.8 Environmental, shelter assemblage.- The First Article shall be subjected to Environmental Testing by the contractor unless otherwise specified by the contract or procurement documents.

Custodian
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Preparing Activity:
Army-EL

Project No. 5410-A201

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-S-49028(EL) SHELTER, ELECTRICAL EQUIPMENT S-538( )/TSQ-84

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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